LECTURE 5 & 6

STRUCTURAL ANALYSIS:PLANAR TRUSSES AND FRAMES

ANALYSIS OF STRUCTURES

- Structural engineering is the field of engineering mostly associated with the design and analysis of engineering structures.

- Engineering structures basically, are designed to support, transmit or modify forces safely. They normally comprise of a number of members connected to form a main structure.

- There are different types of engineering structures, but our focus will be on trusses and frames.

- Trusses and Frames are designed to transmit forces over long distances and comprise straight, **slender** bars that are joined together to form in most cases a pattern of triangles.

- The loads on trusses are always applied in the joints, while in frames, loads may not necessarily be applied at the joints. Also, members of trusses are normally two force members, whereas in frames, at least one member is a multiforce member.

ANALYSIS OF STRUCTURES

Trusses

- Trusses may be classified as simple or compound.

- In simple trusses the number of joints and number of members is related by

$$m = 2j - 3$$

where m is the number of members and j, the number of joints.

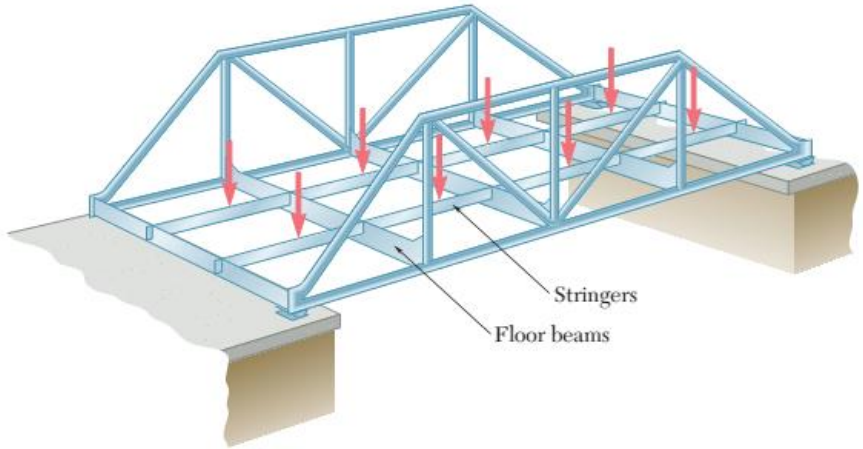
- A truss is therefore often considered simple if its design or construction is such that each time we add 2 new members, they are joined such the number of joints in the truss increases by 1.

- A Truss is designed such that its members are either in tension or compression.

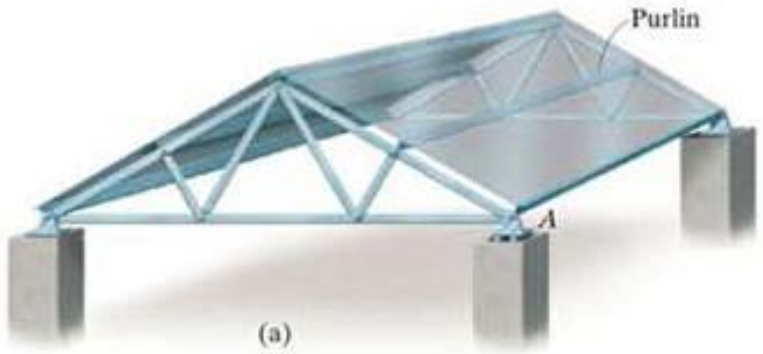
- In designing a truss, it is desirable to know the force each individual member must sustain.

ANALYSIS OF STRUCTURES
Trusses

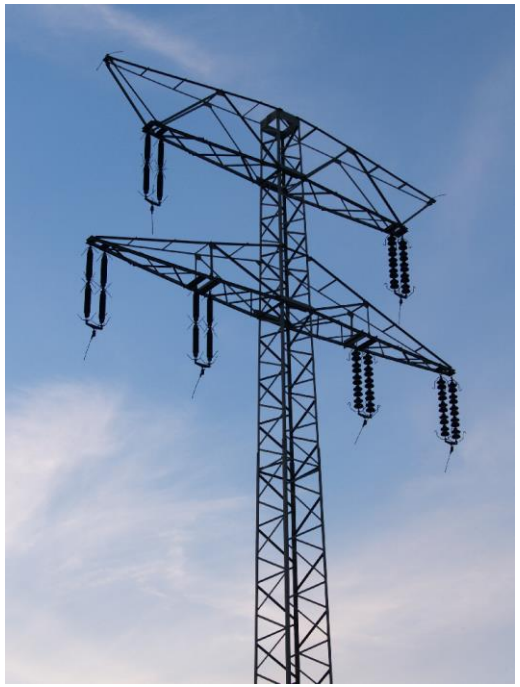
- Common applications are roofs, bridges and power pylons.



Bridge Truss
Source: Mechanics for Engineers by Beet *et al*



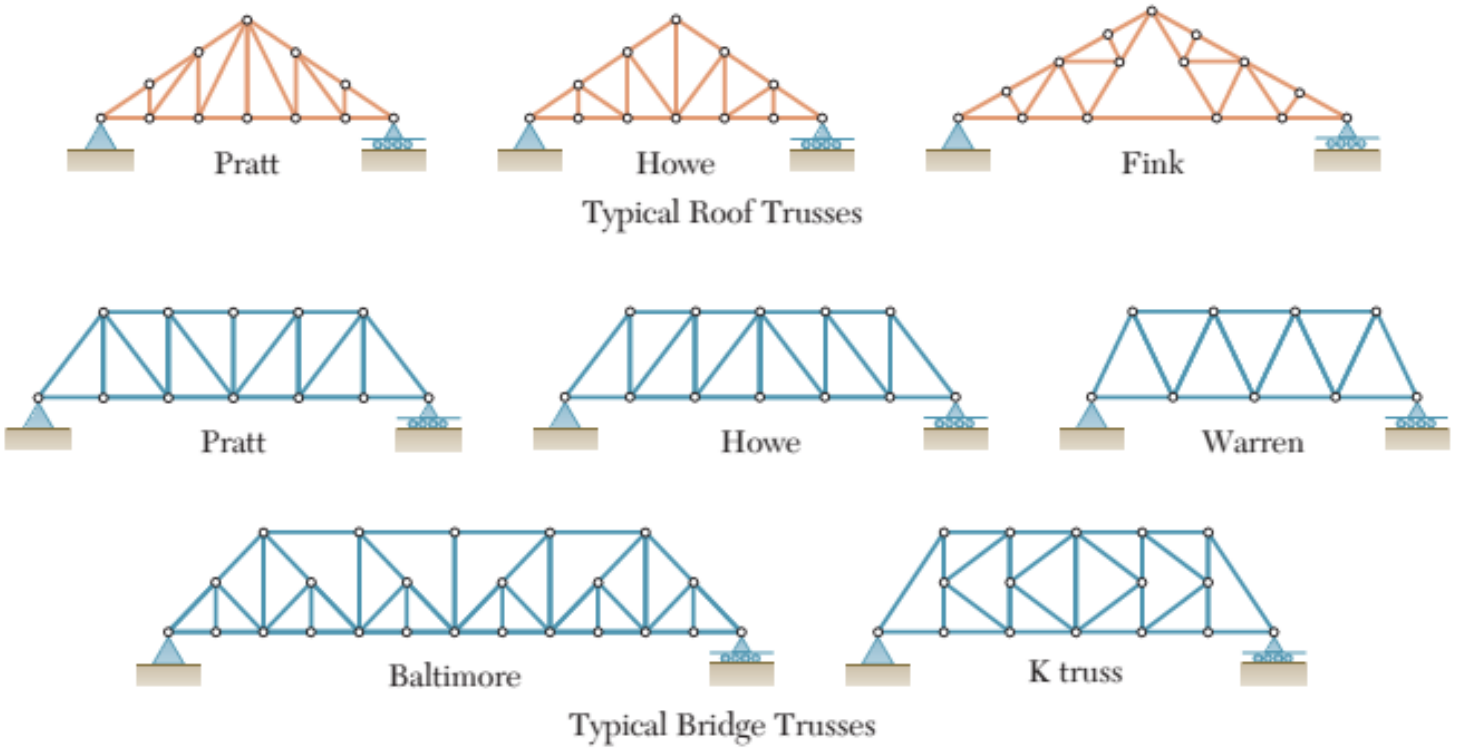
Roof Supporting Truss
Source: Engineering Mechanics Statics by Hiebbler



Power Pylon
Source: http://upload.wikimedia.org/wikipedia/commons/7/7e/Electricity_pylon_power_outage.jpg

ANALYSIS OF STRUCTURES
Planar Trusses

- Trusses may be Planar or Space Trusses.



Some Planar Trusses
Source: Beet *et al*

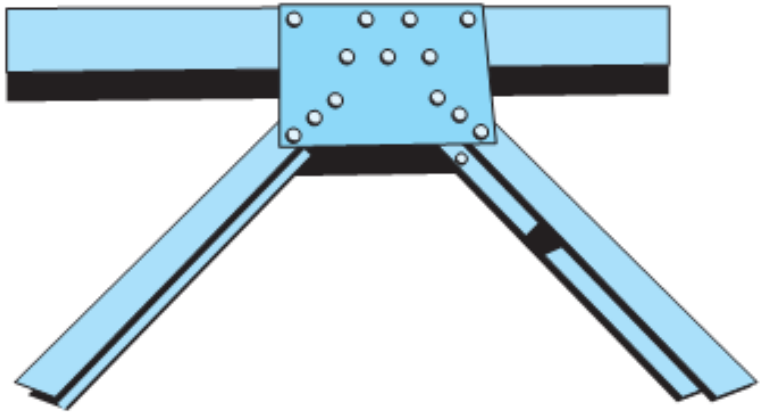


Space Truss
Source
*http://www.picstopin.com/1000/truss/http:%7C%7Cwww*timplex*com%7Ctrusstypess*png/*

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses

- Assumptions for analysis:
  - The weights of the slender bars/members are negligible.
  - Forces act at the ends of the members such that they are in either tension or compression.
  - All joints are pins.

- There are two approaches to analysis:
  - The method of Joints
  - The method of Sections.



Truss members welded or riveted to a gusset plate
Source: Engineering Mechanics – Statics by Pytel

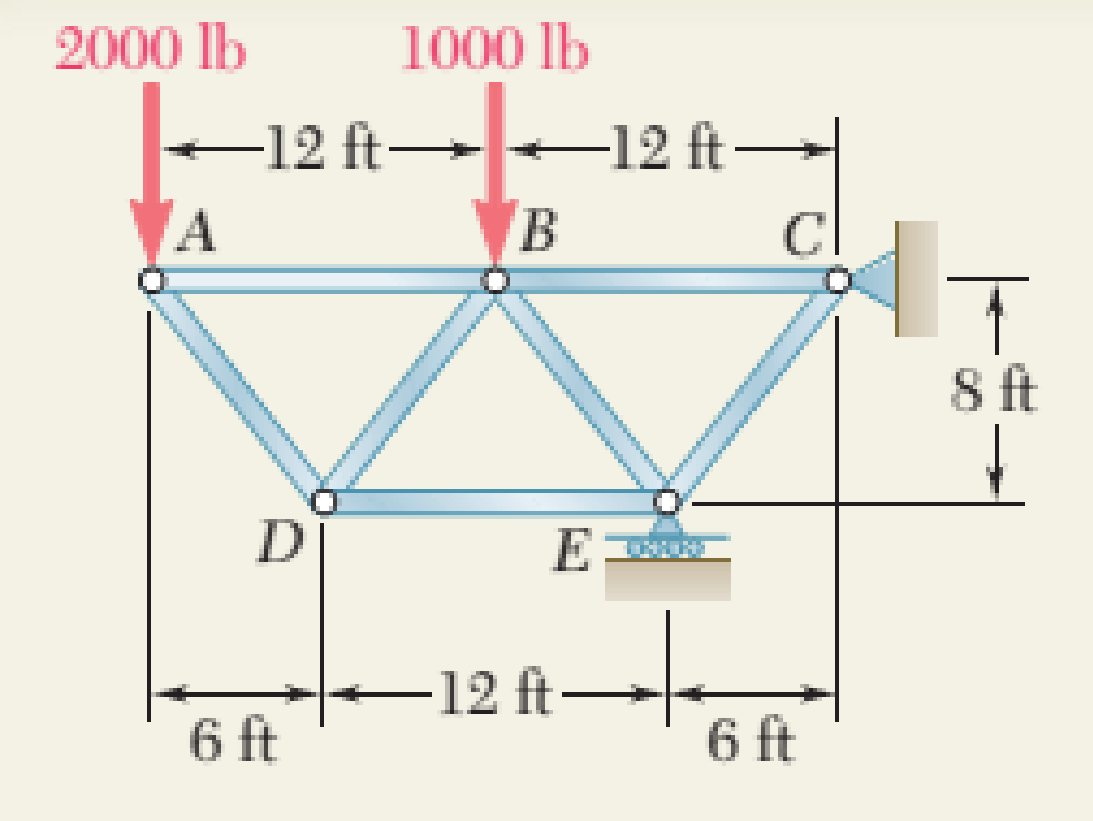
ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

- This involves an equilibrium analysis on each joint to determine the forces being exerted on the end of each member at that joint.
- It is based on the assumption that if the whole truss is in equilibrium, each of its members or joints are also in equilibrium.

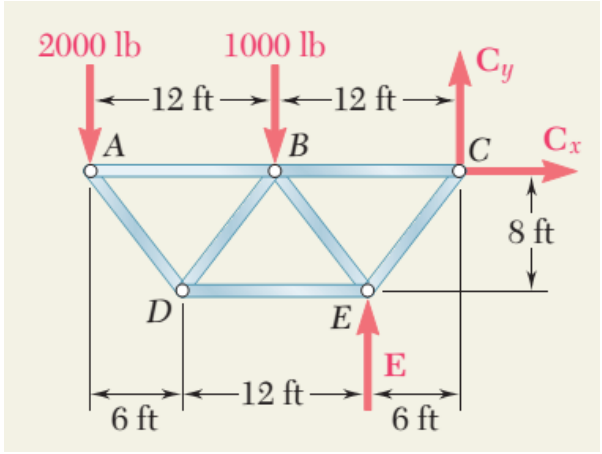
- Analysis is done in two main steps:
  - Determine the reactions at the support reactions using the FBD of the entire truss.
  - Conduct an equilibrium analysis at each joint/pin to determine the forces in each member.

➢Example 5.1

Determine the force in each member of the truss shown below using the method of joints.



- Example 5.1 - Solution
  - Equilibrium analysis on the entire truss to determine the support reactions



For equilibrium,

$$+\rightarrow \sum F_x = 0 : C_x = 0$$

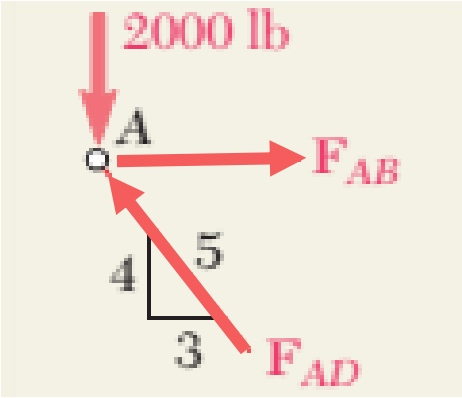
$$+\uparrow \sum F_y = 0 : C_y + E = 3000\,\text{lb}$$

$$\sum M_C = 0\,(\text{clockwise being positive}) : (6\,\text{ft})E - (2000\,\text{lb})(24\,\text{ft}) - (1000\,\text{lb})(12\,\text{ft}) = 0$$

$$E = 10000\,\text{lb}\uparrow$$

  - Equilibrium analysis on the individual joints

Joint A



For equilibriu m,

$$+\rightarrow \sum F_x = 0 : F_{AB} - \frac{3}{5}F_{AD} = 0 \qquad ----(1)$$

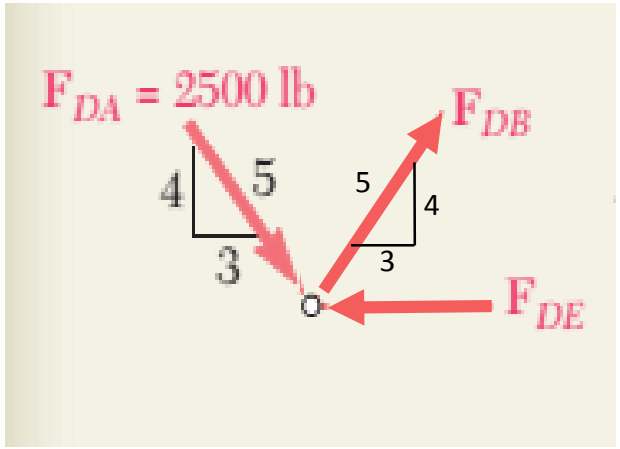
$$+\uparrow \sum F_y = 0 : \frac{4}{5}F_{AD} - 2000\,\text{lb} = 0 \qquad ----(2)$$

Solving (1) and (2) simultaneo usly,

$$F_{AD} = 2500\,\text{lb} \qquad F_{AB} = 1500\,\text{lb}$$

➢Example 5.1 - Solution

Joint D



For equilibrium,

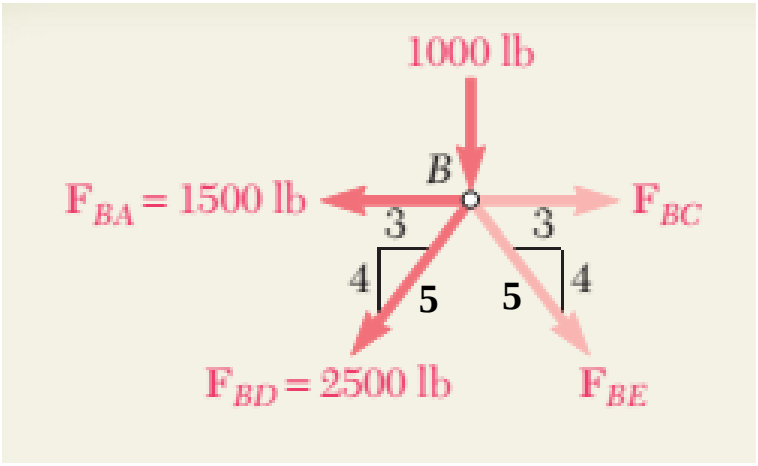
$$+\rightarrow \sum F_x = 0 : \frac{3}{5}(2500\text{lb}) - F_{DE} + \frac{3}{5}F_{DB} = 0 \qquad \text{----(1)}$$

$$+\uparrow \sum F_y = 0 : \frac{4}{5}F_{DB} - \frac{4}{5}(2000\text{lb}) = 0 \qquad \text{----(2)}$$

Solving (1) and (2) simultaneously,

$F_{DB} = 2500\text{lb} \qquad F_{DE} = 3000\text{lb}$

Joint B



For equilibrium,

$$+\rightarrow \sum F_x = 0 : -\frac{3}{5}(2500\text{lb}) - 1500\text{lb} + F_{BC} + \frac{3}{5}F_{BE} = 0 \qquad \text{----(1)}$$

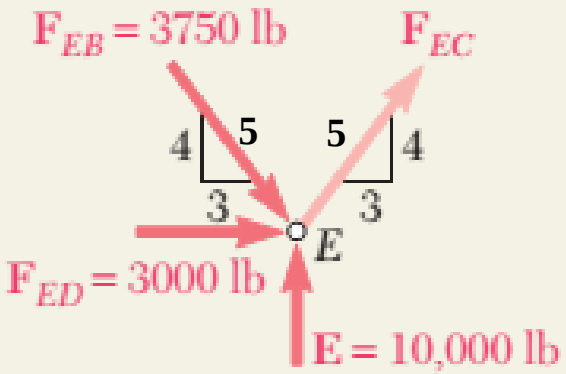
$$+\uparrow \sum F_y = 0 : -1000\text{lb} - \frac{4}{5}(2000\text{lb}) - \frac{4}{5}F_{BE} = 0 \qquad \text{----(2)}$$

Solving (1) and (2) simultaneously,

$F_{BE} = 3750\text{lb} \qquad F_{BC} = 5250\text{lb}$

➢Example 5.1 - Solution

Joint E



For equilibriu m,

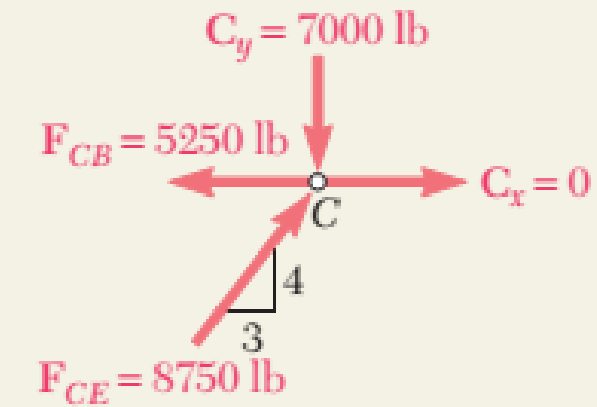
$$+\rightarrow \sum F_x = 0 : \frac{3}{5}(3750\ \text{lb}) + 3000\ \text{lb} + \frac{3}{5}F_{EC} = 0 \qquad \text{----(1)}$$

$$+\uparrow \sum F_y = 0 : -\frac{4}{5}(3750\ \text{lb}) + \frac{4}{5}F_{EC} + 10000\ \text{lb} = 0 \qquad \text{----(2)}$$

Solving (1) and (2) simultaneo usly,

$F_{DB} = 2500\ \text{lb} \qquad F_{DE} = 3000\ \text{lb}$

Joint C



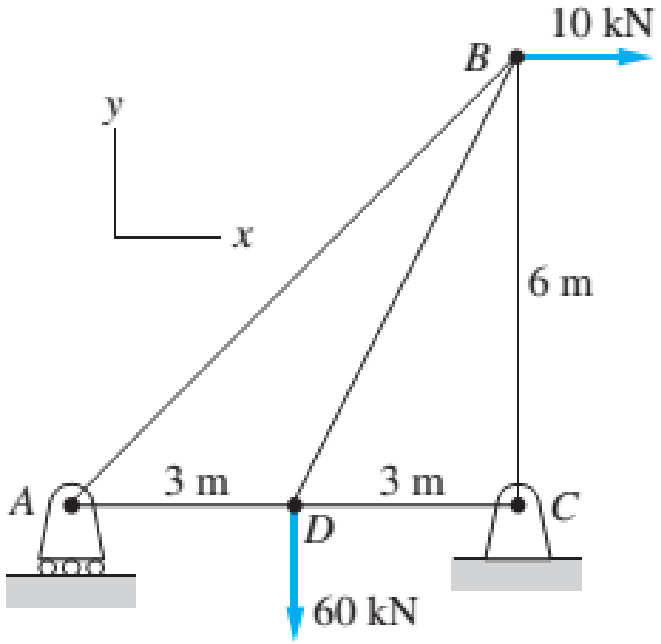
For equilibrium,

$$+\rightarrow \sum F_x = 0 : \frac{3}{5}(8750\,\text{lb}) - 5250\,\text{lb} + 0 = 0$$

$$+\uparrow \sum F_y = 0 : -7000\,\text{lb} - \frac{4}{5}(8750\,\text{lb}) = 0$$

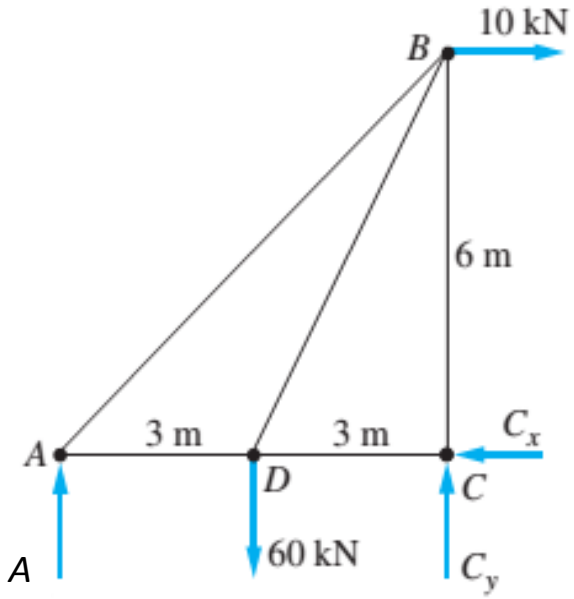
➢Example 5.2

Determine the force in each member of the truss shown below using the method of joints. Indicate whether each member is in tension or compression



➢Example 5.2 - Solution

Equilibrium analysis of the whole structure to determine support reactions



For equilibriu m,

$$+\rightarrow \sum F_x = 0 : -C_x + 10\ \text{kN} = 0$$

$$+\uparrow \sum F_y = 0 : C_y + A - 60\ \text{kN} = 0$$

$$\sum M_C = 0\ (\text{clockwise being positive}) : A(6\ \text{m}) + (10\ \text{kN})(6\ \text{m}) - (60\ \text{kN})(3\ \text{m}) = 0$$

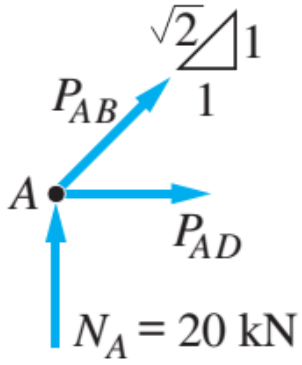
$$A = 20\ \text{kN}$$

$$\therefore C_y = 40\ \text{kN}$$

➢Example 5.2 - Solution

Equilibrium analysis on the joints

Joint A



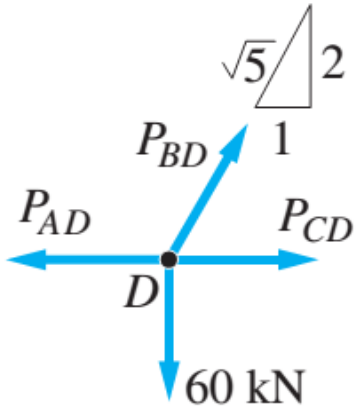
For equilibrium,

$$+\rightarrow \sum F_x = 0 : P_{AD} + \frac{1}{\sqrt{2}} P_{AB} = 0 \qquad \text{----- (1)}$$

$$+\uparrow \sum F_y = 0 : 20\,\text{kN} + \frac{1}{\sqrt{2}} P_{AB} = 0 \qquad \text{------(2)}$$

$P_{AB} = -28.3\,\text{kN}$ (Compression) $\qquad P_{AD} = 20\,\text{kN}$ (Tension)

Joint D



For equilibriu m,

$$+\rightarrow \sum F_x = 0 : -P_{AD} + \frac{1}{\sqrt{5}} P_{BD} + P_{CD} = 0 \qquad \text{----- (1)}$$

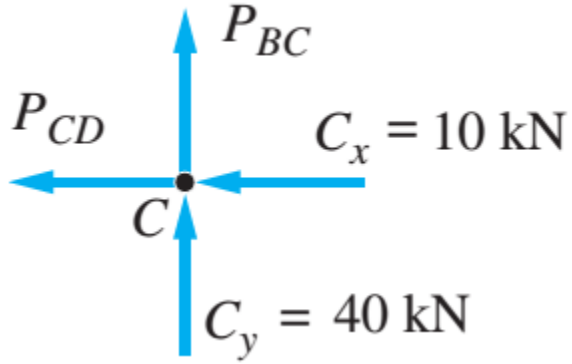
$$+\uparrow \sum F_y = 0 : -60\,\text{kN} + \frac{2}{\sqrt{5}} P_{BD} = 0 \qquad \text{------(2)}$$

$P_{BD} = 67.1\,\text{kN}$ (Tension) $\qquad P_{CD} = -10\,\text{kN}$ (Compression)

➢Example 5.2 - Solution

Equilibrium analysis on the joints

Joint C



For equilibriu m,

$$+\rightarrow \sum F_x = 0 : -P_{CD} - 10\ \text{kN} = 0 \qquad \text{-----} \ (1)$$

$$+\uparrow \sum F_y = 0 : 40\ \text{kN} + P_{BC} = 0 \qquad \text{------} (2)$$

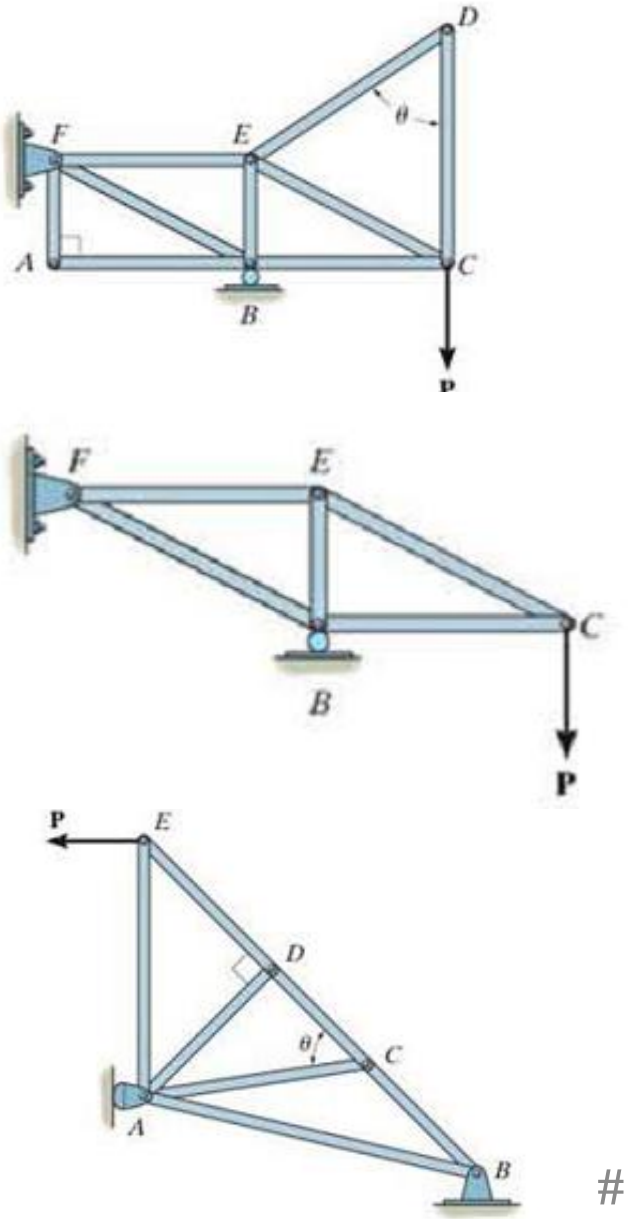
$$P_{BC} = -40\ \text{kN (Tension)}$$

ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Joints

Zero Force Members

- It is possible during analysis to realize during analysis that some members do not experience any external forces. Such members are referred to as ***Zero Force Members.***
- Identification of such members can significantly make analysis of planar trusses using the method of joints easier.
- Identification is normally by inspection.
- However, two rules of thumb for identifying such members are;

1. *if three members from a truss joint for which two of the members are collinear (have the same line of action), the third member may be a zero force member provided no external force or support reaction is applied to the joint.*
2. *If a joint is formed by only two members and the joint is not subjected to any external load or support reactions, then the two members may be zero force members.*

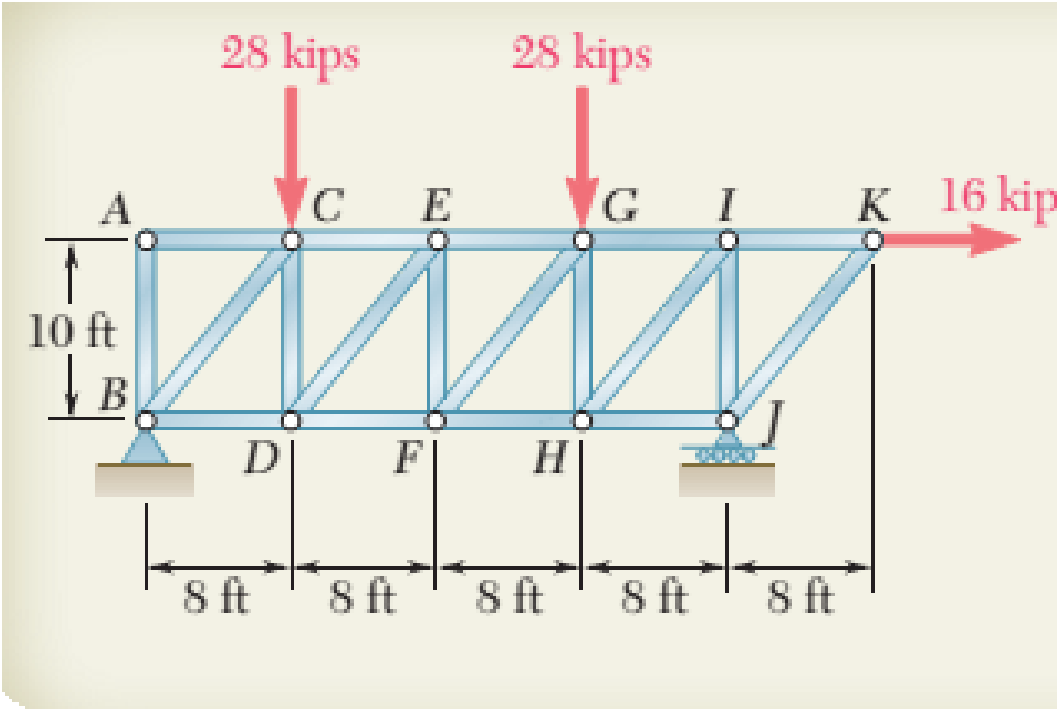


ANALYSIS OF STRUCTURES
Analysis of Planar Trusses - The Method of Sections

- This method is useful if it is desired to find the forces in only few members.
- It is based on the assumption that if the whole truss is in equilibrium, any section of it must also be in equilibrium.
- Analysis in the following steps:
  - Determine the support reactions using the FBD of the entire truss.
  - Draw a line which divides the truss into two separate portions but does not intersect more than three members. One of the three intersected members must be the member of interest.
  - Use one of the two portions as a free body diagram and determine the force due to the member of interest.

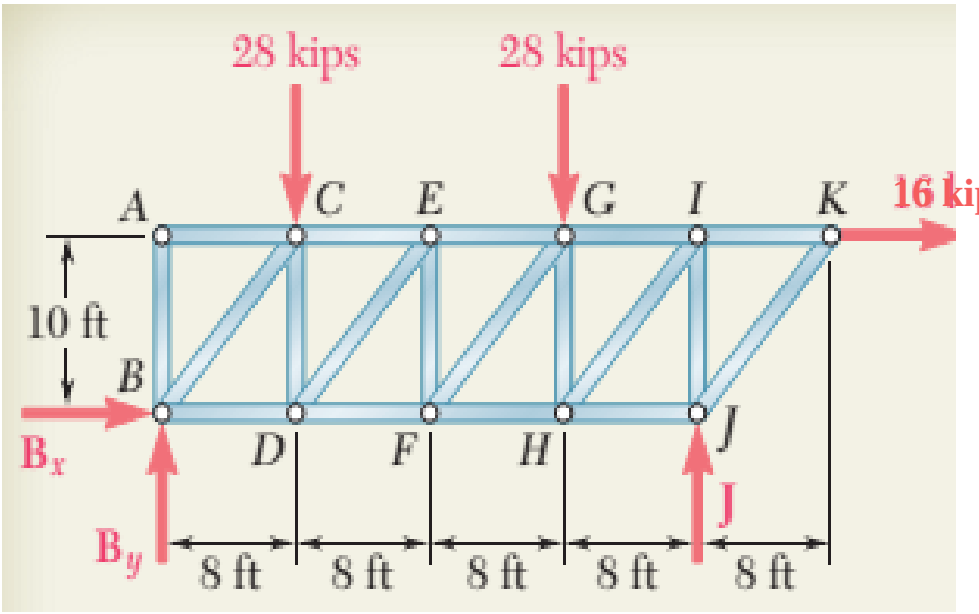
➢Example 5.3

Determine the force in members *EF* and *GI* of the truss shown



➢Example 5.3 - Solution

Equilibrium analysis on the whole truss to determine support reactions.



For equilibriu m,

$$+\rightarrow \sum F_x = 0 : B_x + 16 \text{ kips} = 0$$

$$+\uparrow \sum F_y = 0 : B_y + J - 28 \text{ kips} - 28 \text{ kips} = 0$$

$$\sum M_B = 0 \text{ (clockwise being positive)} :$$

$$28 \text{ kips}(8 \text{ ft}) + 28 \text{ kips}(24 \text{ ft}) + 16 \text{ kips}(10 \text{ ft}) - J(32 \text{ ft}) = 0$$

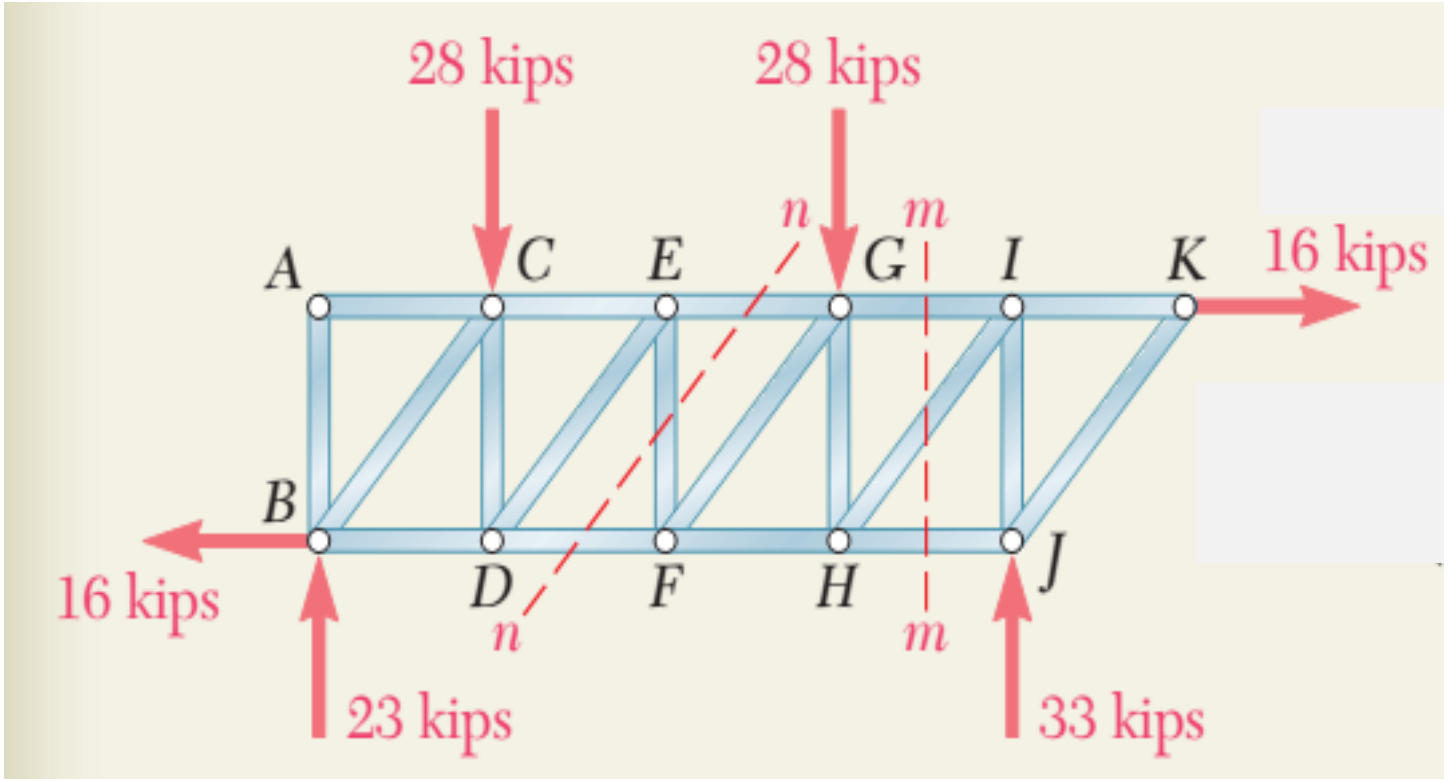
$$J = 33 \text{ kips}$$

$$B_x = -16 \text{ kips}$$

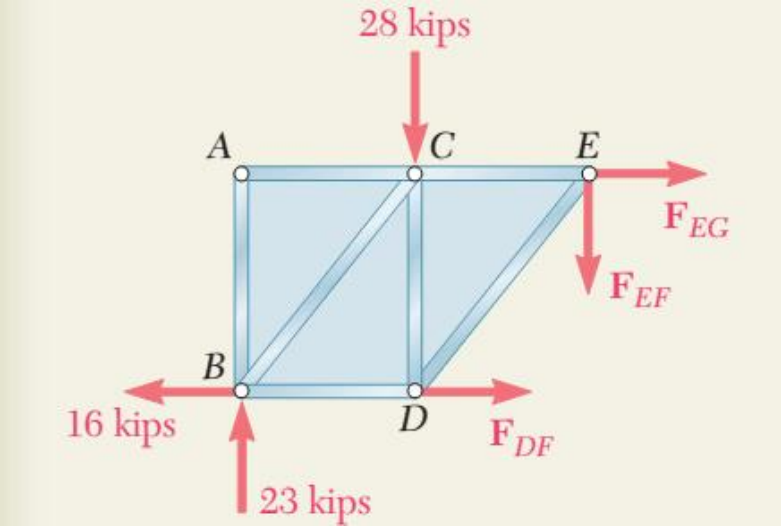
$$B_y = 23 \text{ kips}$$

➢Example 5.3 - Solution

Dividing the truss into two with line *nn* (to solve for member *EF*), then with *mm* (to solve for *GI*)



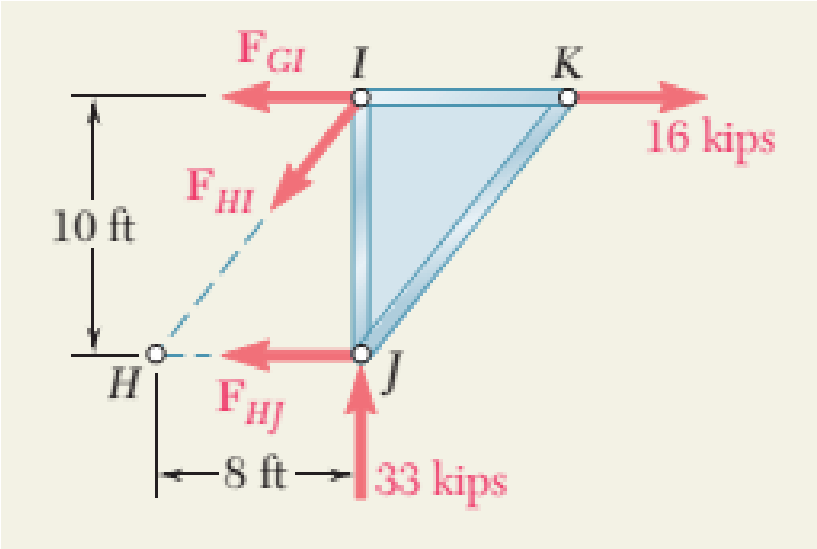
➢Example 5.3 - Solution



Performing an equilibriu m analysis on the left side of *nn* to obtain *EF*,

$$+\uparrow \sum F_y = 0 : 23\,\text{kips} - 28\,\text{kips} - F_{EF} = 0$$

$$F_{EF} = -5\,\text{kips}$$



Performing an equilibriu m analysis on the right side of *mm* to obtain *GI*,

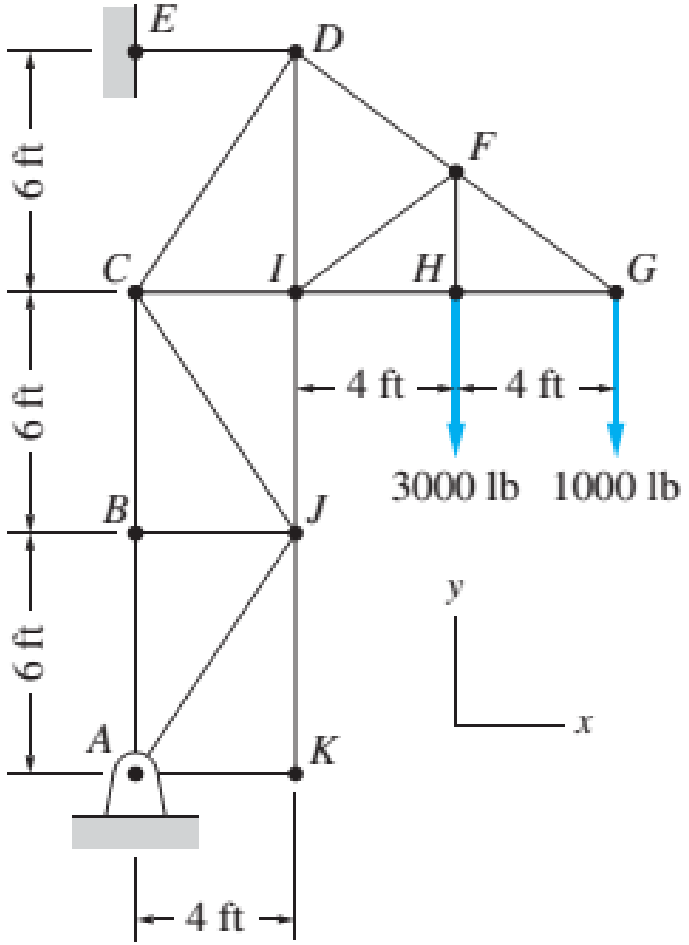
$$\sum M_H = 0\,(\text{clockwise being positive}) :$$

$$-F_{GI}(10\,\text{ft}) - 33\,\text{kips}(8\,\text{ft}) + 16\,\text{kips}(10\,\text{ft}) = 0$$

$$F_{GI} = -10.4\,\text{kips}$$

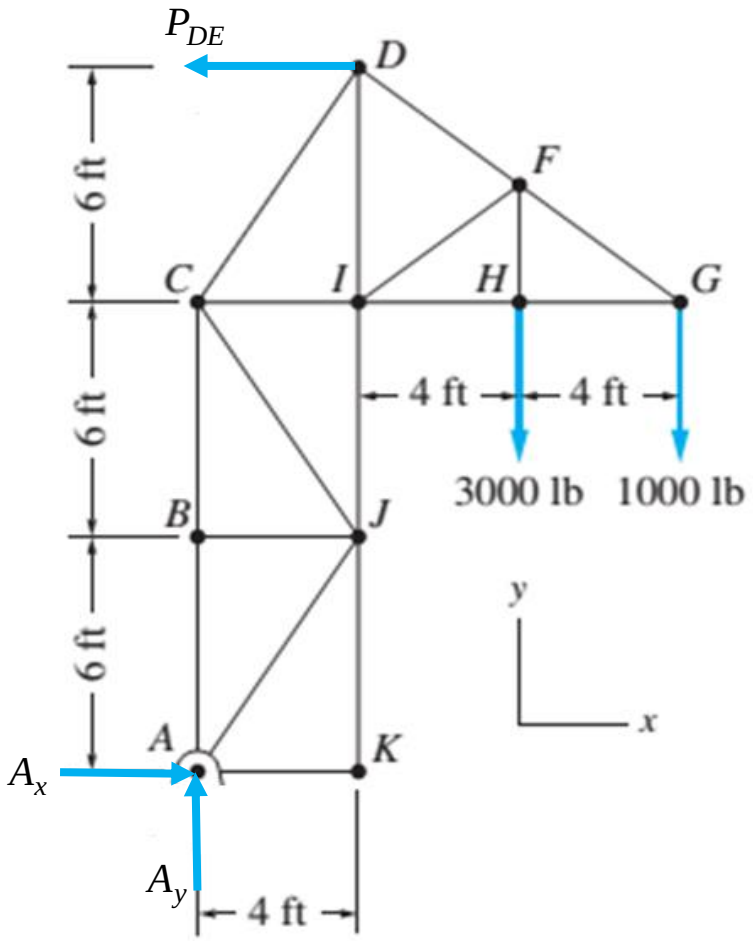
➢Example 5.4

Determine the forces in members *FI* and *JC* of the truss shown. Indicate tension or compression.



➢Example 5.4 - Solution

Equilibrium analysis on the whole truss to determine support reactions gives.



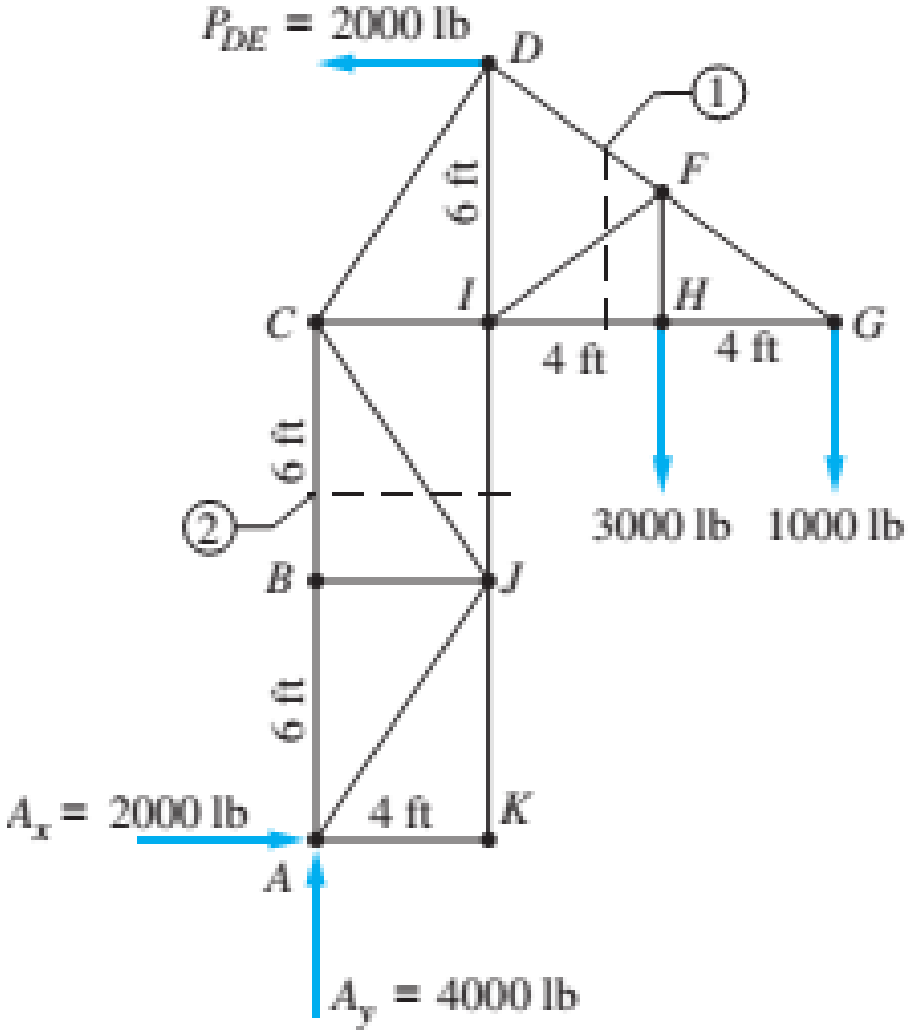
$$P_{DE} = 2000 \text{ lb}$$

$$A_x = 2000 \text{ lb}$$

$$A_y = 4000 \text{ lb}$$

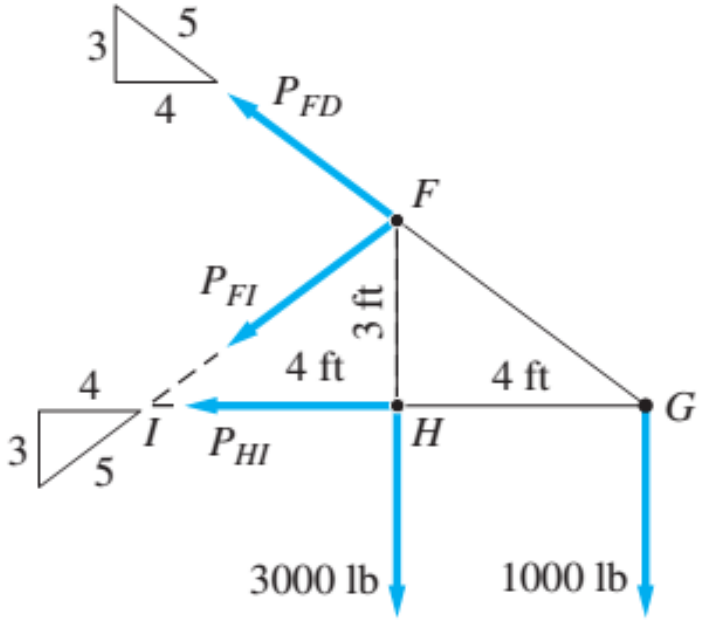
➢Example 5.4 - Solution

The truss is divided into 2 with line 1 so *FI* can be determined, and with line 2 so *JC* can be determined.



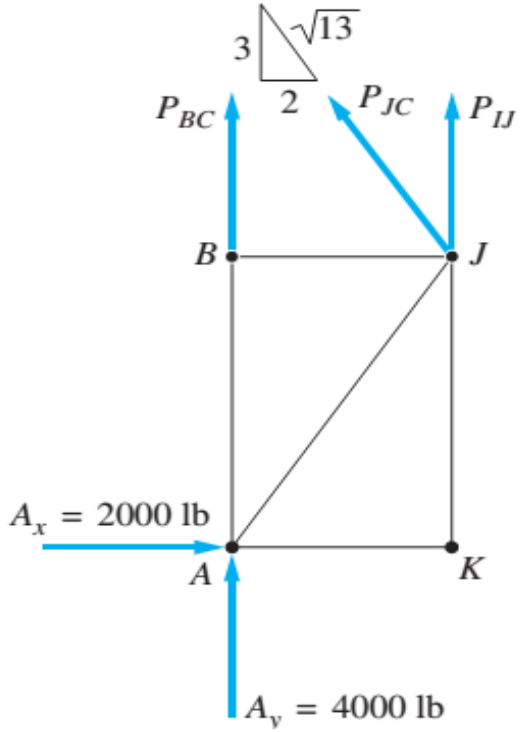
➢Example 5.4 - Solution

For *FI*



$$P_{FI} = -2500 \text{ lb}$$

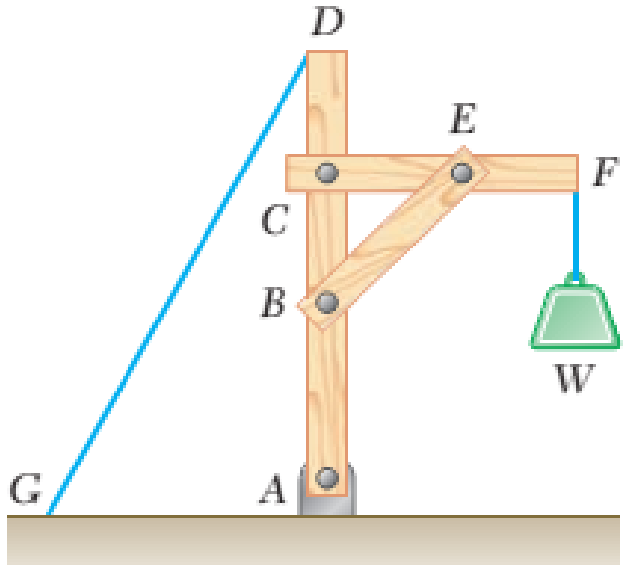
For *JC*

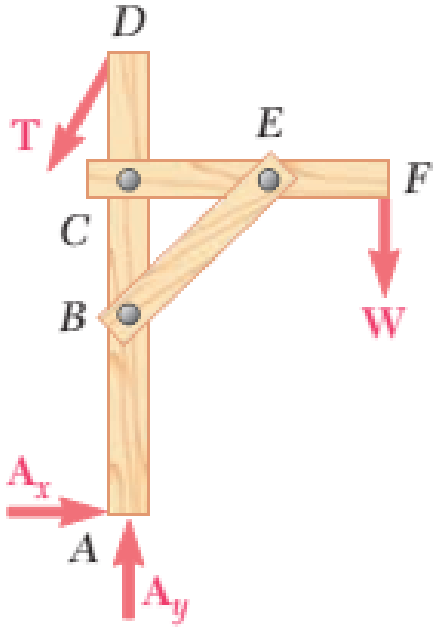


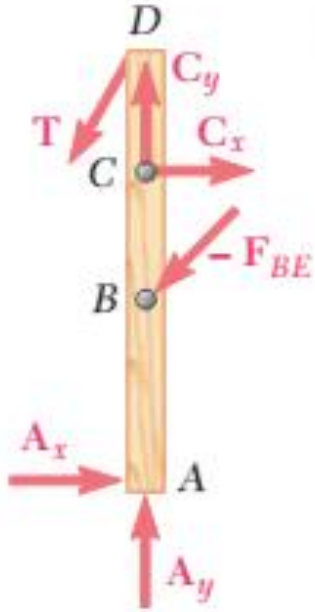
$$P_{JC} = -3610 \text{ lb}$$

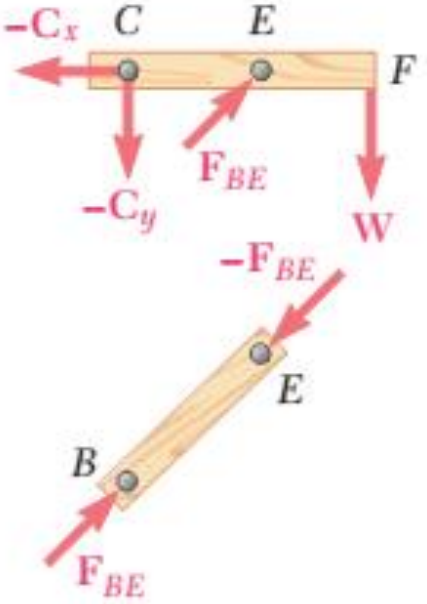
ANALYSIS OF STRUCTURES
Frames

- Frames and Machines are structures in which at least one member is a multiforce member (acted upon by three or more forces).
- While frames are designed to support forces, machines transmit and modify forces.



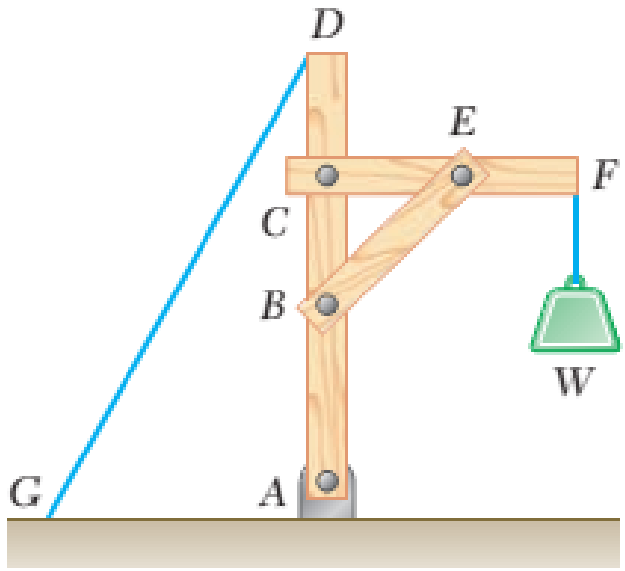


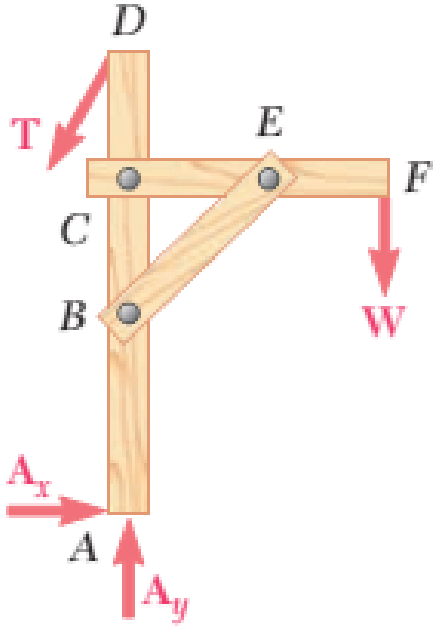


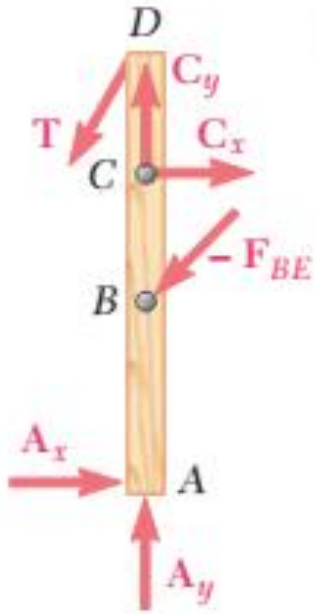


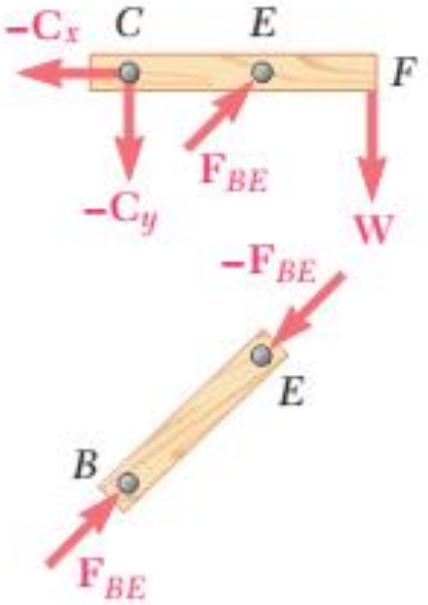
ANALYSIS OF STRUCTURES
Analysis of Frames

- Procedure is similar to what is used in trusses.
- An equilibrium analysis is first carried out on the entire frame.
- An equilibrium is then carried out on the individual members of the frame to determine all forces acting on each of them (It is sometimes easier if this is done first on the two force, then three force members).



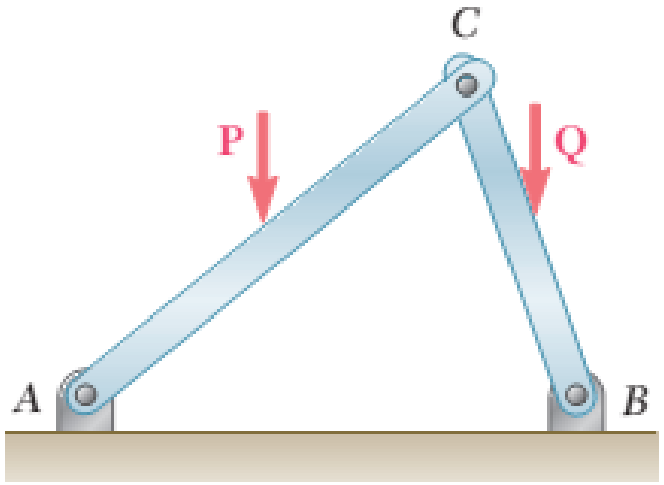


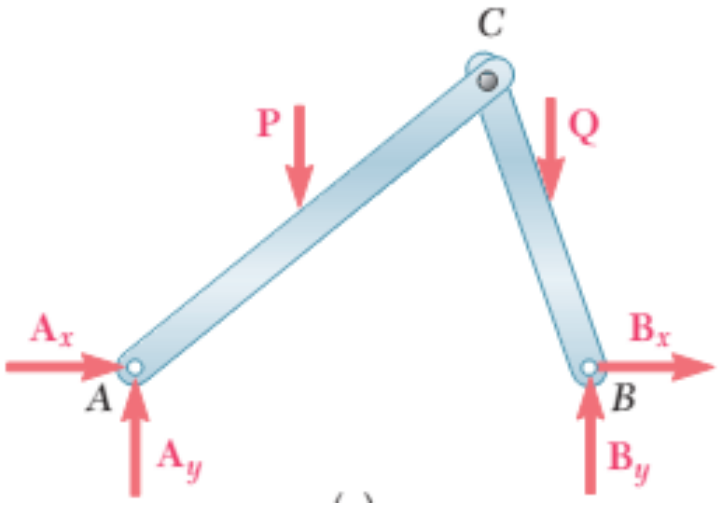


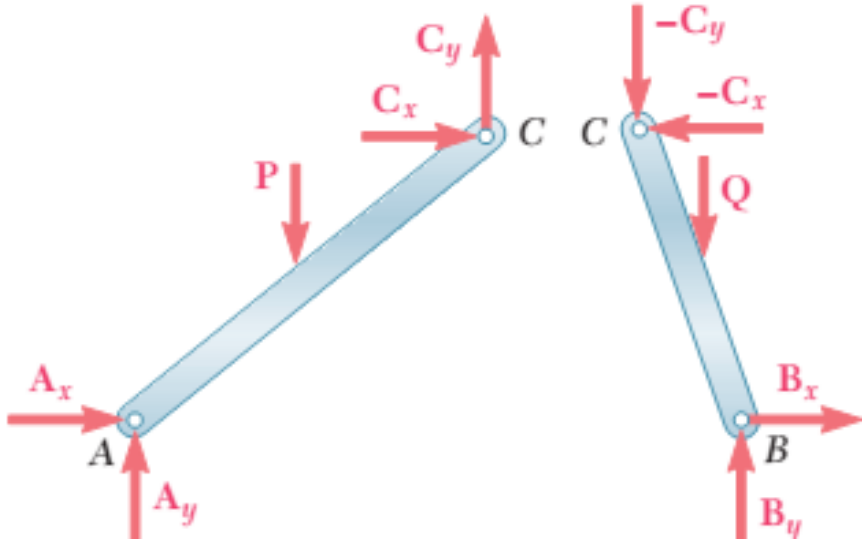


ANALYSIS OF STRUCTURES
Analysis of Frames

- During analysis of non-rigid frames, the individual members of frame are considered as rigid members.

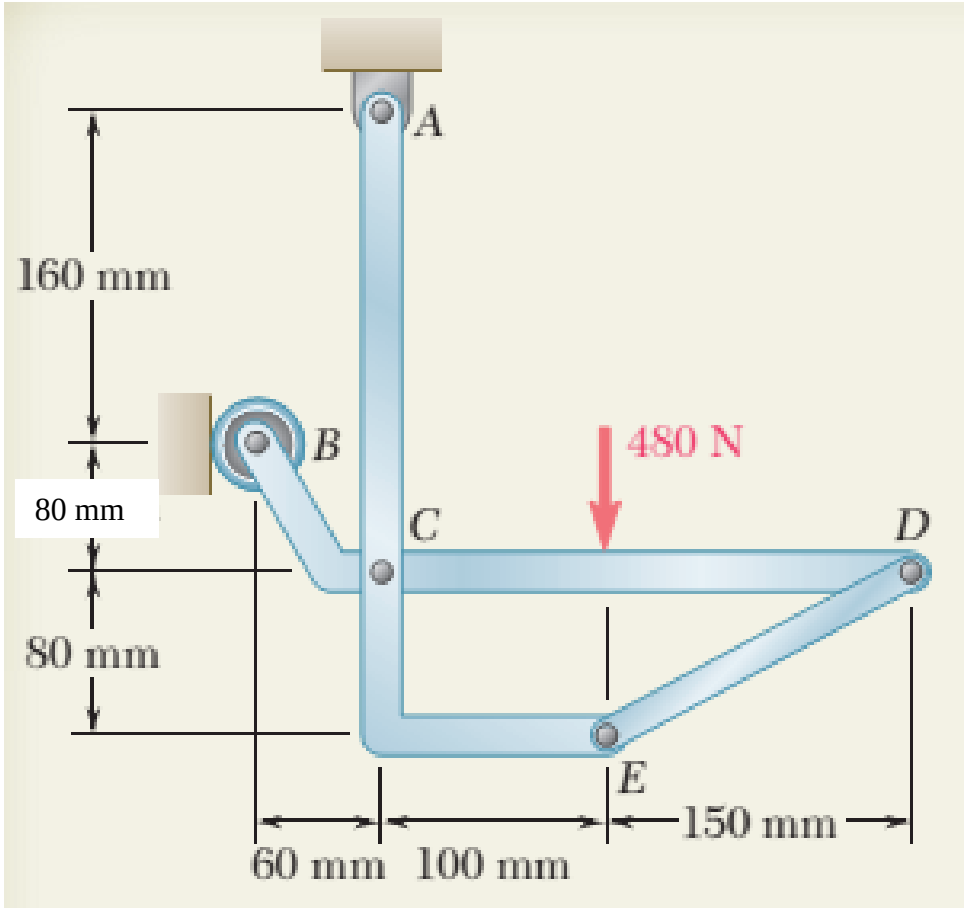






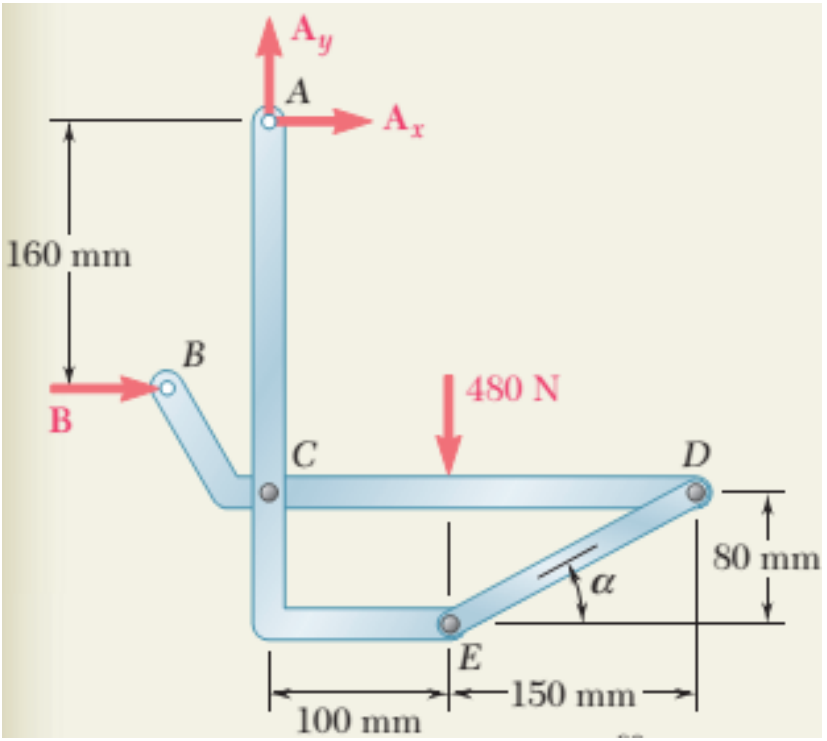
➢Example 5.5

In the frame shown, members *ACE* and *BCD* are connected by a pin at *C* and by the link *DE*. For the loading shown, determine the force in link *DE* and the components of the force exerted at *C* on member *BCD*. Is link *DE* tension or compression?



➢Example 5.5 – Solution

Equilibrium analysis on the entire structure



For equilibrium,

$$+\rightarrow \sum F_x = 0 : B + A_x = 0$$

$$+\uparrow \sum F_y = 0 : A_y = 480 \text{ N}$$

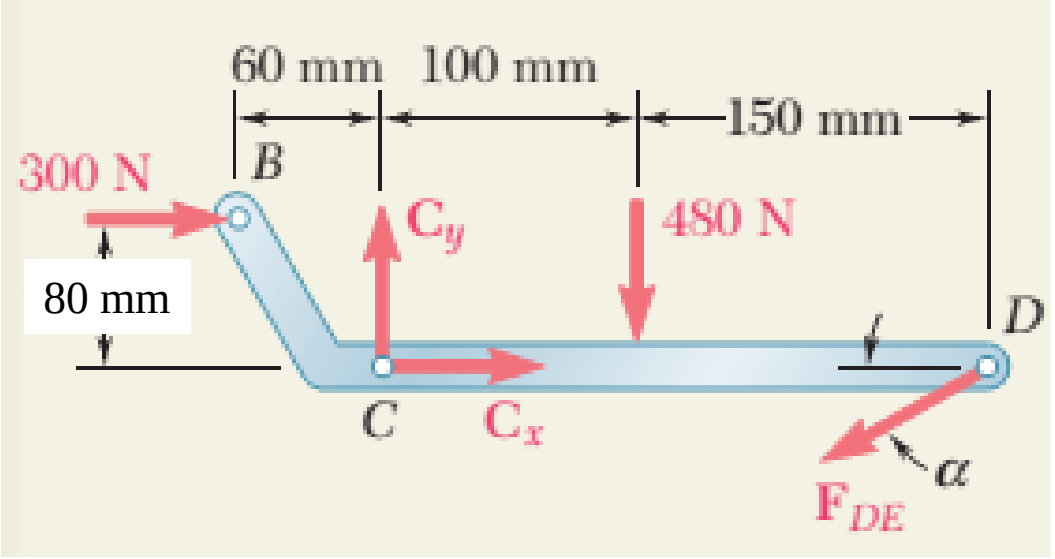
$$+\circlearrowleft \sum M_A = 0 : -B(160 \text{ mm}) + 480 \text{ N}(100 \text{ mm}) = 0$$

$$B = 300 \text{ N}$$

So,

$$A_x = -300 \text{ N}$$

Example 5.5 – Solution (Equilibrium analysis on individual links)



For equilibrium,

$$+\rightarrow \sum F_x = 0 : 300\ \text{N} + C_x - F_{DEx} = 0$$

$$+\uparrow \sum F_y = 0 : C_y - 480\ \text{N} - F_{DEy} = 0$$

$$+\circlearrowleft \sum M_C = 0 : 300\ \text{N}(80\ \text{mm}) + 480\ \text{N}(100\ \text{mm}) + F_{DEy}(250\ \text{mm}) = 0$$

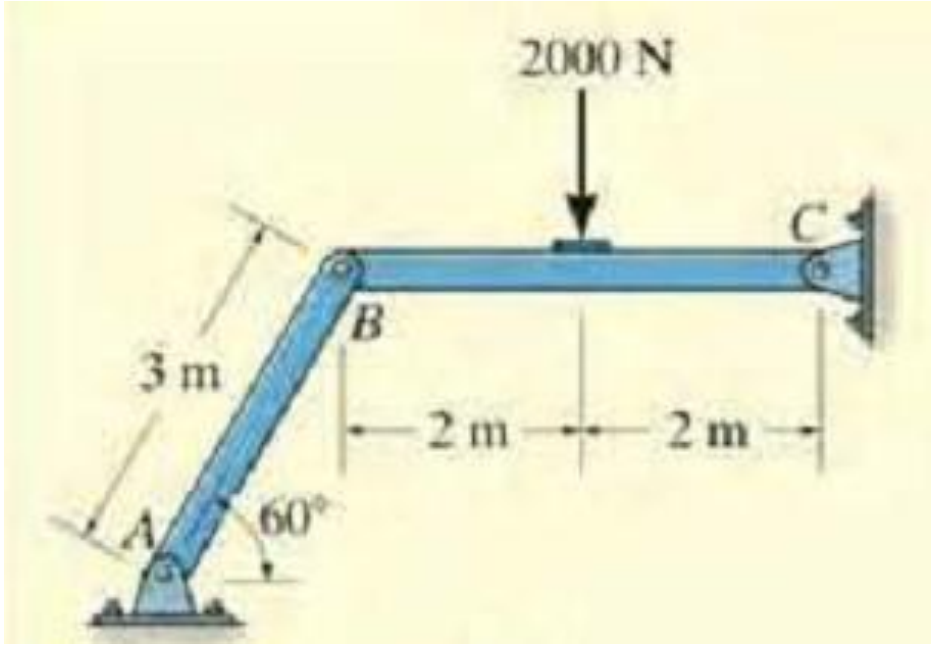
$$F_{DEy} = -264\ \text{N} \qquad C_y = 216\ \text{N}$$

$$\alpha = \tan^{-1}\left(\frac{80}{150}\right)$$

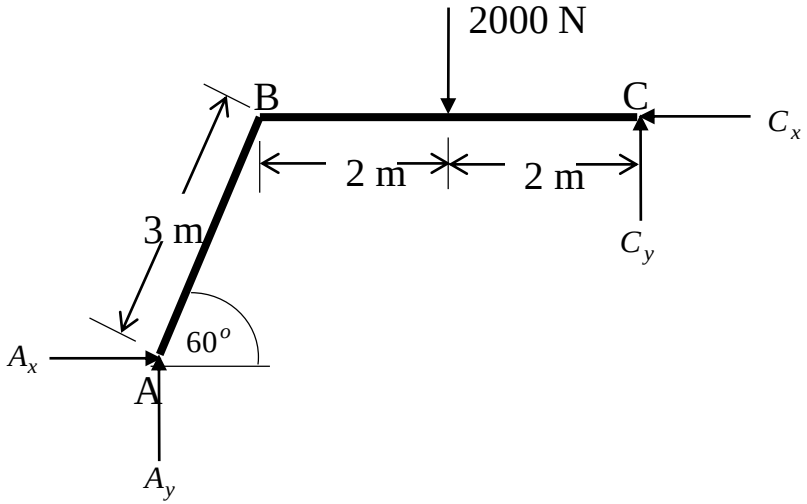
$$F_{DE} = -561\ \text{N} \qquad F_{DEx} = -495\ \text{N} \qquad C_x = -795\ \text{N}$$

➢Example 5.6

Determine the horizontal and vertical components of the force which the pin at *C* exerts on member *BC* of the frame shown below. Also determine the forces acting on both members at *B*



Example 5.6 - Solution



For equilibriu m,

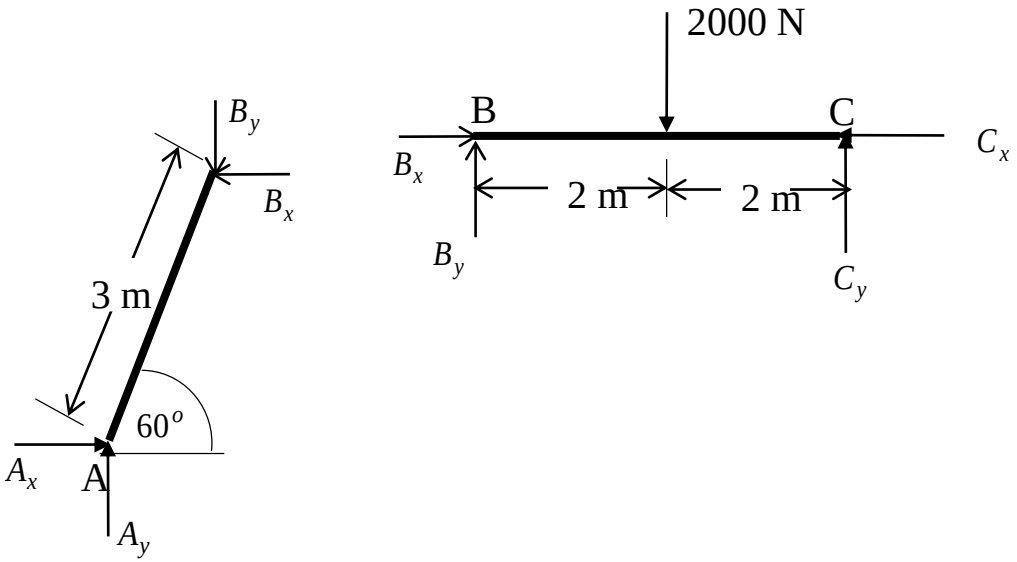
$$+\rightarrow \sum F_x = 0 : A_x - C_x = 0$$

$$+\uparrow \sum F_y = 0 : A_y + C_y - 2000\ \text{N} = 0$$

$$+\circlearrowleft \sum M_C = 0 : -A_y(4 + 3\cos 60^o) + A_x\left(3\sin 60^o\right) + 2000\ \text{N}(2\ \text{m}) = 0$$

$$+\circlearrowleft \sum M_A = 0 : -C_y(4 + 3\cos 60^o)\text{m} + C_x\left(3\sin 60^o\right)\text{m} + 2000\ \text{N}\left(2 + 3\cos 60^o\right)\text{m} = 0$$

$$A_x = C_x = 577\ \text{N} \qquad A_y = C_y = 1000\ \text{N}$$



Taking member BC, for equilibriu m,

$$+\rightarrow \sum F_x = 0 : B_x - C_x = 0$$

$$+\uparrow \sum F_y = 0 : B_y + C_y - 2000\ \text{N} = 0$$

$$B_x = 577\ \text{N}$$

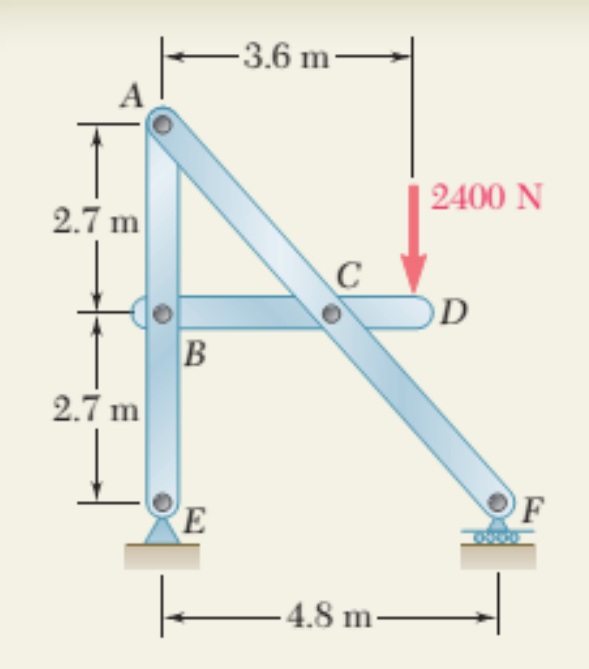
$$B_y = 1000\ \text{N}$$

For member AB,

$$B_x = 577\ \text{N} \leftarrow$$

$$B_y = 1000\ \text{N} \downarrow$$

- Example 5.7
- Determine the components of the forces acting on each member of the frame shown.



$$F = 1800 \text{ N}$$
$$E_y = 600 \text{ N}$$
$$E_x = 0 \text{ N}$$
$$A_x = 0 \text{ N}$$
$$A_y = 1800 \text{ N}$$
$$B_x = 0 \text{ N}$$
$$B_y = 1200 \text{ N}$$
$$C_x = 0 \text{ N}$$
$$C_y = 1000 \text{ N}$$

LECTURE 7

A BRIEF OVERVIEW FRICTION

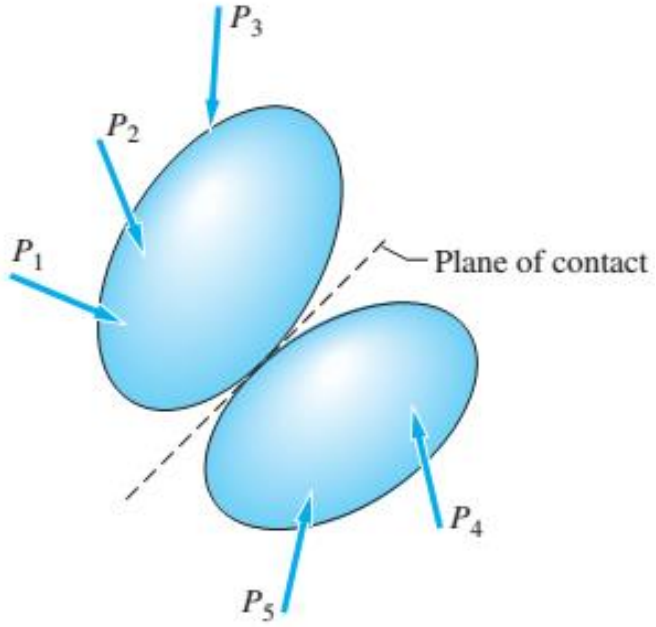
FRICTION

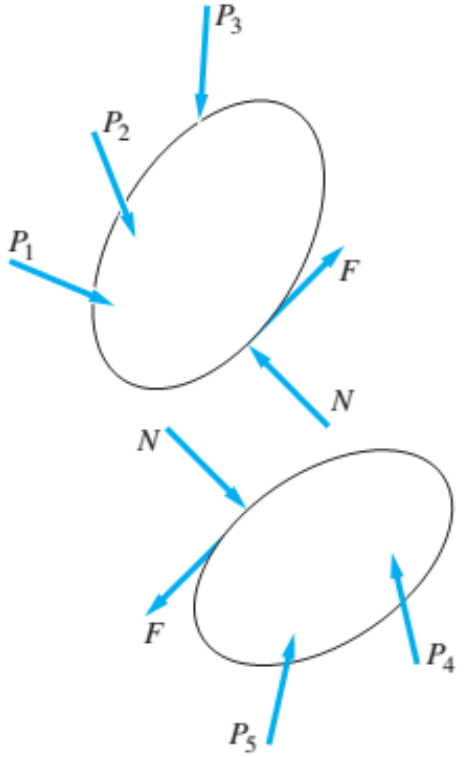
- Friction resists relative sliding motion between two surfaces that are in contact.
- Friction may be helpful; as seen during walking, applications to clutches, belts etc.
- It may also be detrimental; wear in machinery, reduction of efficiency in transmission of power by converting mechanical energy into heat.
- It plays a very important role in a lot of engineering applications.

- Friction may be dry or wet.
- *Dry* or *Coloumb* Friction occurs when there is no layer of fluid separating the contacting surfaces.

FRICTION
Coloumb's Theory of Dry Friction

- Has been found to give satisfactory results in practical problems.
- Explanation:
  - Two bodies in contact, will each experience a normal reaction, N and Friction force, F which will act in a direction opposite to that the body wants to move in.





FRICTION

Coloumb's Theory of Dry Friction

- The relation between the Normal and the Friction force can be one of three cases:
  - Static Case: where there is no sliding movement between the two bodies, N and F are related by

$$F \leq F_{\max} = \mu_s N$$

and there will be no movement so long as

$$F \leq F_{\max}$$

  - Impending Sliding Case:

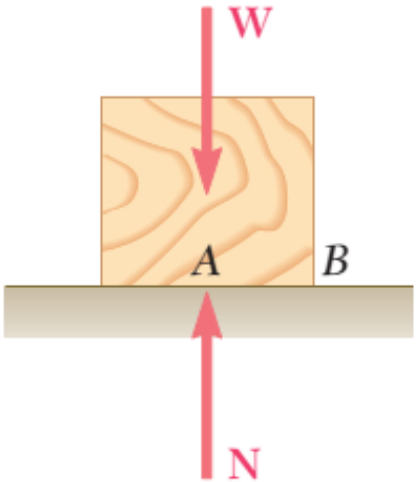
$$F = F_{\max} = \mu_s N$$

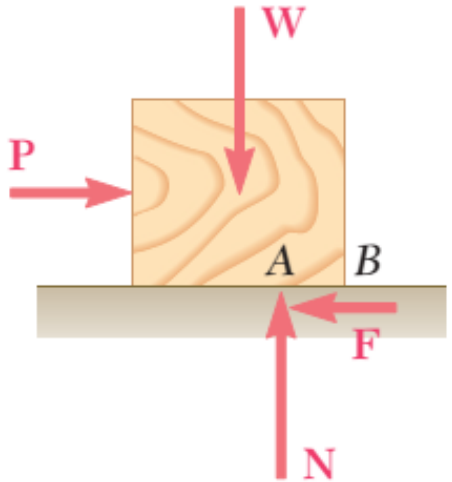
  - Dynamic Case: when the two bodies are already sliding relative to each other,

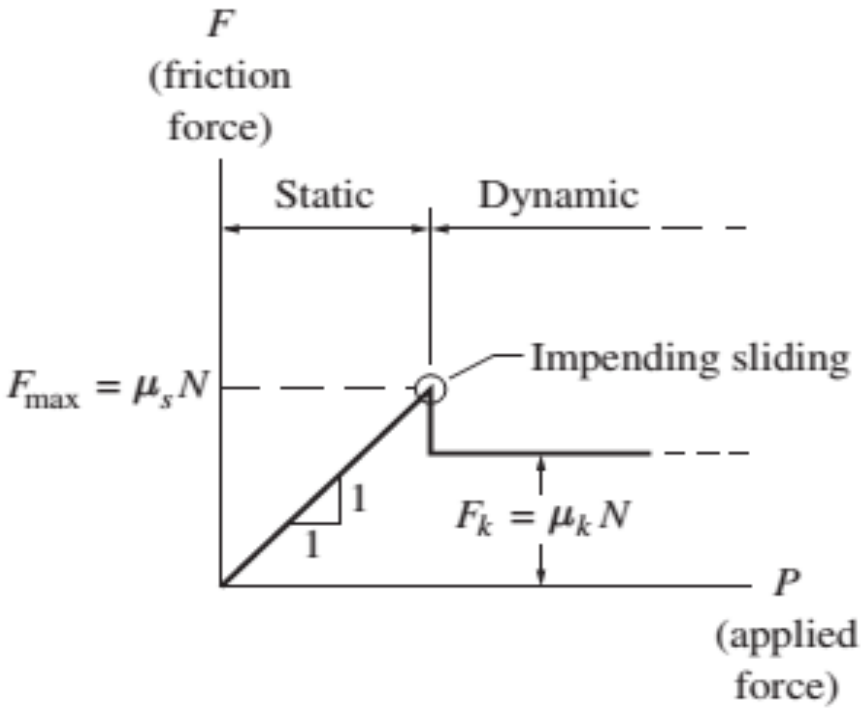
$$F_k = \mu_k N$$

FRICTION
Coloumb's Theory of Dry Friction

- When a force P is applied to a body in the direction shown, the body experiences friction in the direction shown. The body will move when P exceeds F.
- Once the body is moving, the force P required to keep it moving is less than what was required to make it start moving. Hence, Static Friction Force > Dynamic Friction force





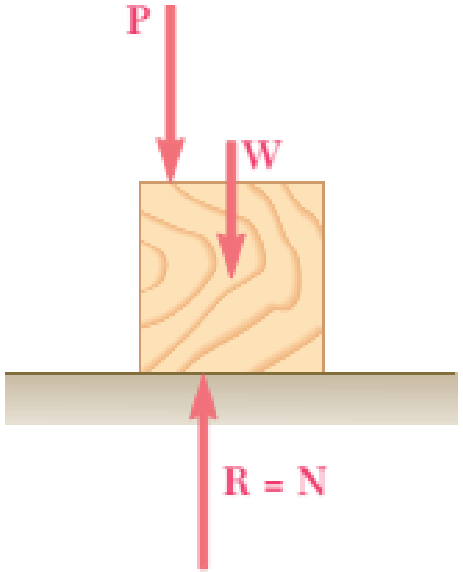


Variation of Friction Force with Applied Force
Source: Engineering Mechanics – Statics by Pytel

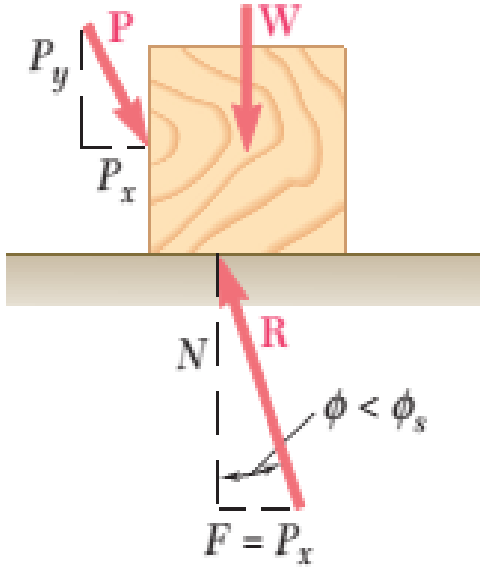
FRICTION

Angles of Dry Friction

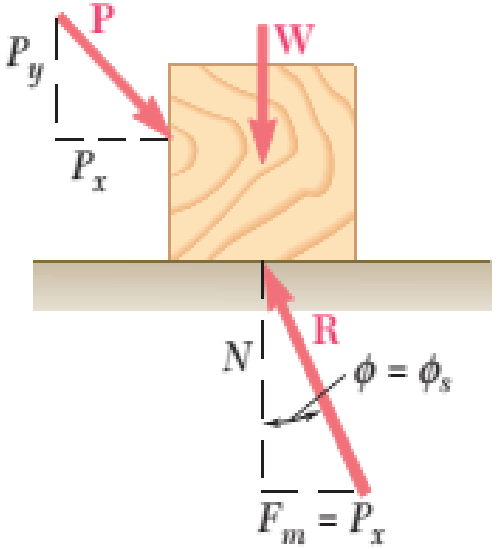
- The Friction force and Normal Reaction acting on a body can be resolved into a resultant.
- The angle this resultant makes with the normal reaction is referred to as the angle of friction.
- There can be the angle of static friction and the angle of kinetic friction.



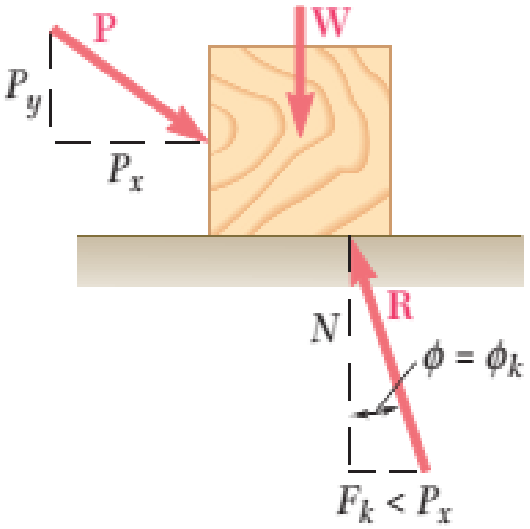
(a) No friction



(b) No motion



(c) Motion impending



(d) Motion

$$\tan\phi_s = \frac{F_m}{N} = \frac{\mu_s N}{N}$$

$$\tan\phi_s = \mu_s$$

$$\tan\phi_k = \frac{F_k}{N} = \frac{\mu_k N}{N}$$

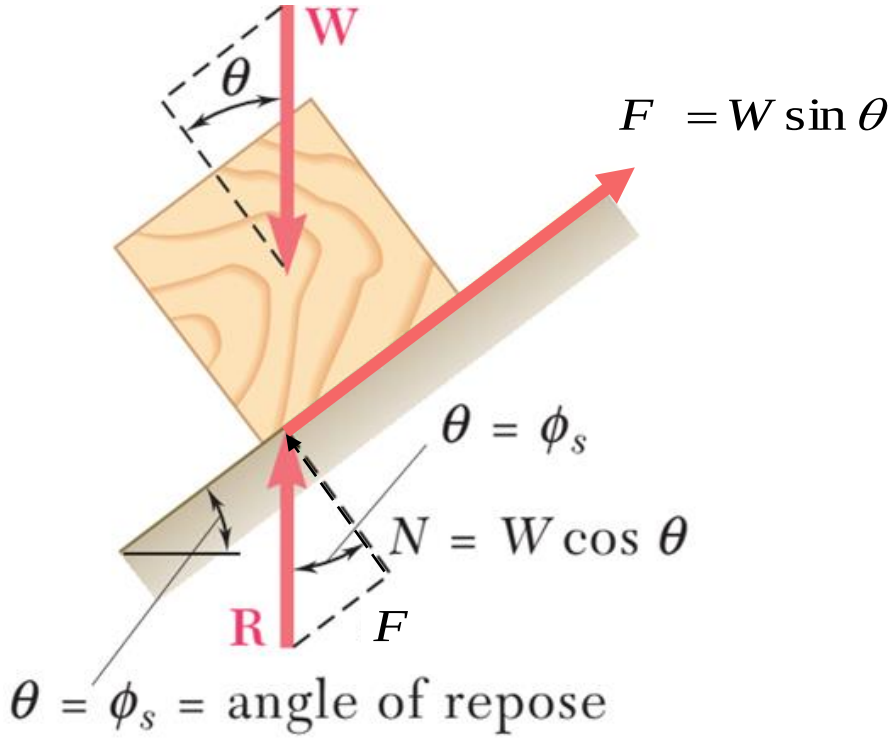
$$\tan\phi_k = \mu_k$$

FRICTION
Angles of Dry Friction

- The co-efficient of static friction between two surfaces is determined experimentally.
- A block of one material is placed on a plane of variable inclination of the second material.
- Inclination is increased until sliding is imminent, in which case $F_s = F_{\max} = \mu_s N$

$$\phi_s = \theta = \frac{F}{N} = \frac{\mu_s N}{N} = \mu_s$$

- θ is known as the *angle of repose*
- $\mu_s = \tan\theta$



Motion impending

FRICTION
Laws of Dry Friction

- Coloumb and other subsequent works, give the following laws of friction:
  - The maximum frictional resistance that can be developed is independent of the size of the areas in contact.
  - The maximum static Friction Force, *Fmax* and the dynamic friction force F_k are each proportional to the normal reaction between the surfaces.

$$F_{\max} = \mu_s N$$

$$F_k = \mu_k N$$

where μ_s and μ_k are the static and kinetic co-efficients of friction respectively.

  - The limiting value of Static friction force is greater than the kinetic friction force
  - For low velocities, the kinetic frictional resistance is practically independent of velocity.

FRICTION
Some Limitations

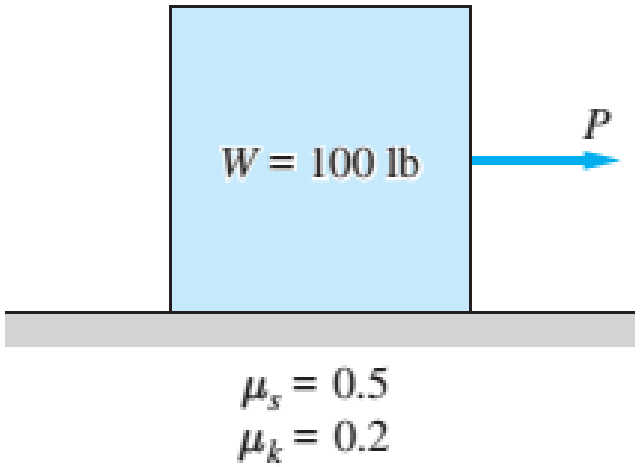
- Some limitations of Coloumb's Theory and other related works include:
  - The co-efficients of friction are all determined experimentally and are at best approximations. They may vary with environmental conditions, the condition of the surfaces among other factors.
  - The theory of dry friction is applicable only to surfaces that are dry or that contain only a small amount of lubricant. If there is relative motion between the surfaces of contact, the theory is valid for low speeds only.
  - There are situations where the amount of friction between surfaces depends on the area of contact. For example, the traction (friction force) between an automobile tire and the pavement can be increased under certain conditions by letting a small amount of air out of the tire, thus increasing the contact area.

FRICTION

- Problems involving dry friction are solved in pretty much the same way as equilibrium problems.
- Free body diagrams on which forces and dimensions must be correctly indicated.
- Friction forces must always oppose the direction of motion.
- Problems involving dry friction may fall into one of three possible scenarios;
  - It is not known if there is impending sliding
    - Assume equilibrium, solve equilibrium equations for unknowns, check to see if $F \leq \mu_s N$. If not, the problem is a dynamic problem in which there is sliding.
  - There is impending sliding, and impending sliding surfaces are known.
    - $F = F_{\max} = \mu_s N$, solve the equilibriums equations for the unknowns.
  - There is impending sliding, but the impending sliding surfaces are not known.
    - Identify surfaces for possible motion, set $F = F_{\max} = \mu_s N$, solve equilibrium equations for unknowns and choose correct surface by comparison of results.

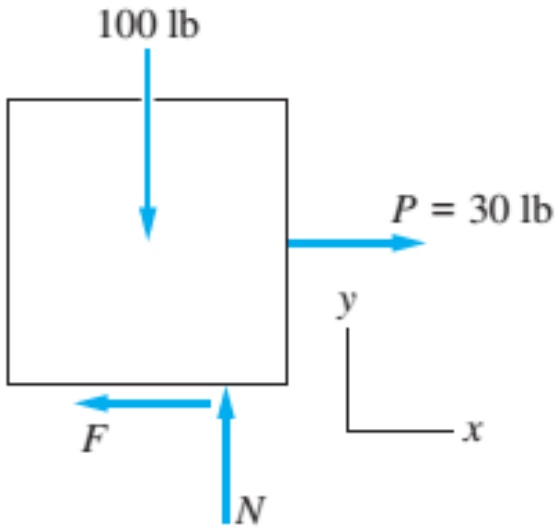
FRICTION

- Example 7.1
- A force of P of 30 lb is applied as shown to a 100 lb block that was initially at rest. Determine if the block will slide or not.



Example 7.1 – Solution

FBD



Assuming equilibriu m,

$$+\rightarrow \sum F_x = 0; P - F = 0$$

$$+\uparrow \sum F_y = 0; -100\ \text{lb} + N = 0$$

$$F = 30\ \text{lb}$$

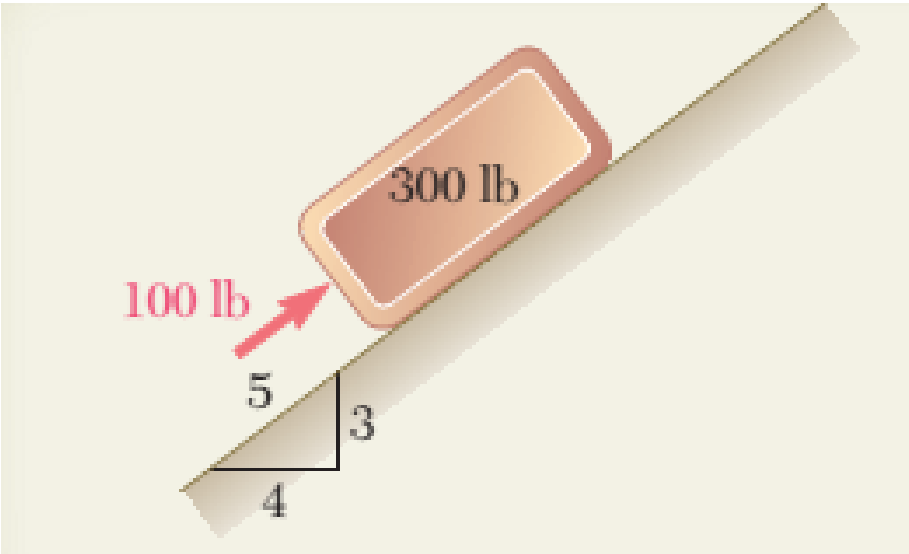
$$N = 100\ \text{lb}$$

$$F_{\max} = \mu_s N = 0.5(100\ \text{lb}) = 50\ \text{lb}$$

Since $F < F_{\max}$, the body is in static equilibriu m and will remain at rest.

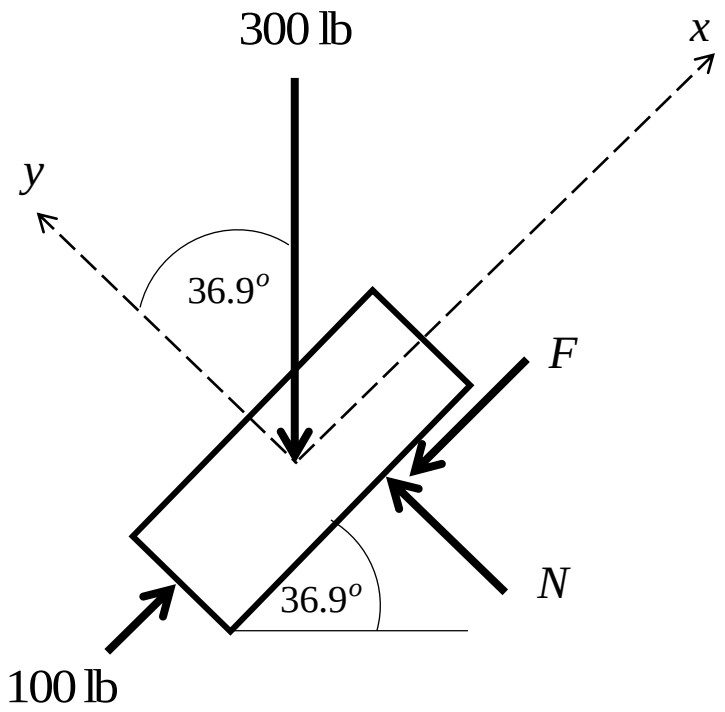
FRICTION

- Example 7.2
- A 100-lb force acts as shown on a 300-lb block placed on an inclined plane. The coefficients of friction between the block and the plane are $\mu_s = 0.25$ and $\mu_k = 0.20$. Determine if the block will slide down the plane and also determine value of the friction force.



FRICTION

➢Example 7.2 – Solution z



Assuming equilibriu m,

$$+\nearrow \sum F_x = 0; 100\text{ lb} - (300 \sin 36.9)\text{lb} - F = 0$$

$$+\nwarrow \sum F_y = 0; -(300 \cos 36.9)\text{lb} + N = 0$$

$$F = -80\text{ lb}$$

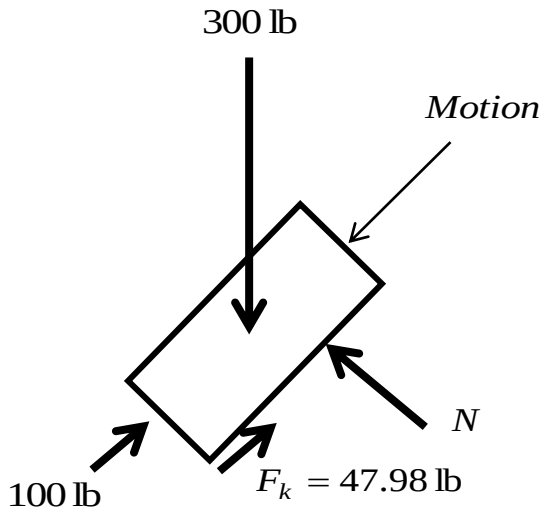
$$N = 239.9\text{ lb}$$

$$F_{\max} = \mu_s N = 0.25(239.9\text{ lb}) = 59.98\text{ lb}$$

Since $|F| > |F_{\max}|$, the body will not be at rest. It will slide down the plane.

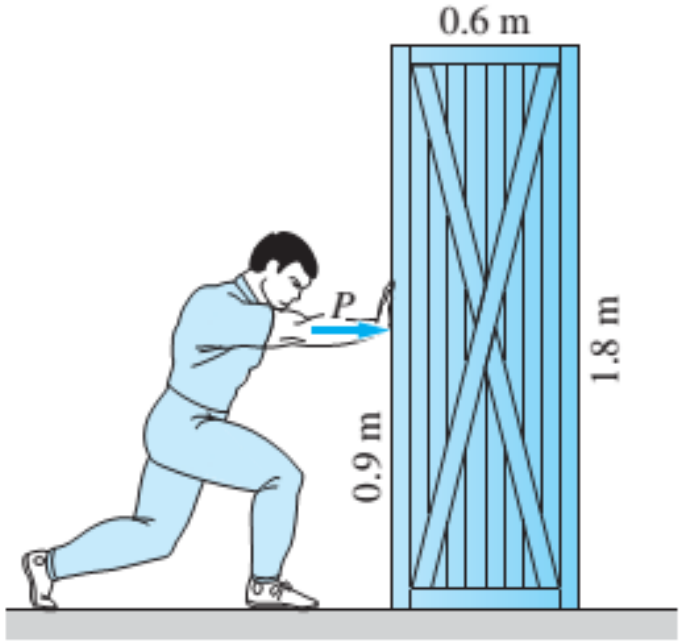
Therefore, Force due to friction is dynamic. It will be given by

$$F_k = \mu_k N = 0.20(239.9\text{ lb}) = 47.98\text{ lb}$$

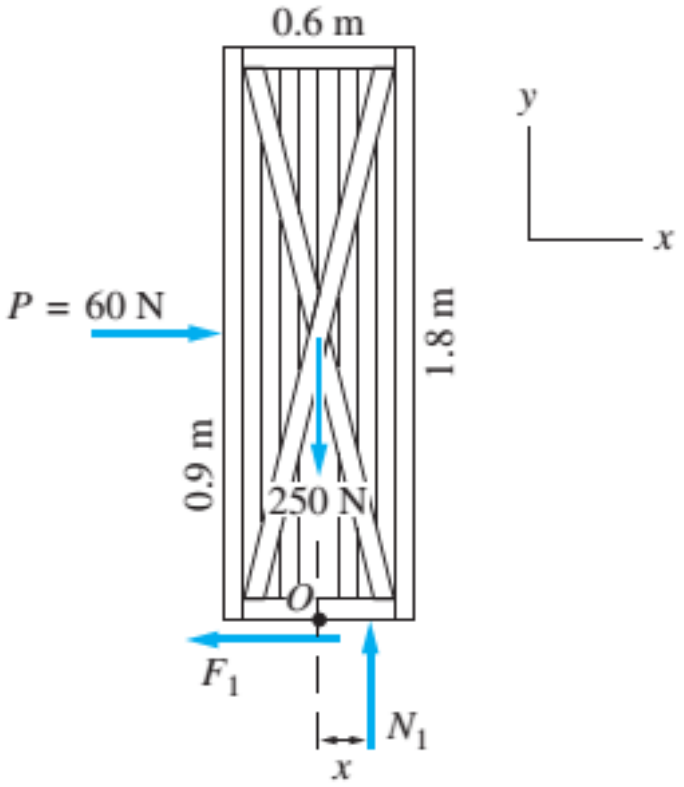


FRICTION

- Example 7.3
- A force of 60 N is being applied to a 250 N crate by a man as shown below. Determine if the crate will remain in static equilibrium. The weight of the crate acts through its geometric center, and the coefficient of static friction between the crate and the floor is 0.3



➢Example 7.3 - Solution



Assuming equilibrium,

$$+\rightarrow \sum F_x = 0; \; -F_1 + 60\ \text{N} = 0$$

$$+\uparrow \sum F_y = 0; \; N_1 - 250\ \text{N} = 0$$

$$+\circlearrowleft \sum M_O = 0; \; -N_1 x + (P)0.9\ \text{m} = 0$$

$x = 0.216\ \text{m} < 0.3\ \text{m}$

The crate will not tip over

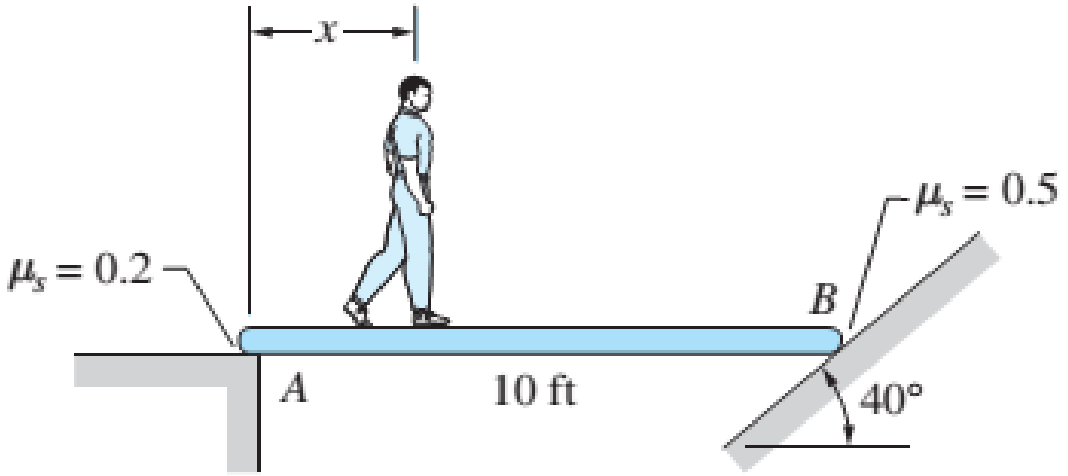
$F_{1\max} = \mu_s N_1 = 0.3(250) = 75\ \text{N}$

$\text{P} < F_{1\max}$

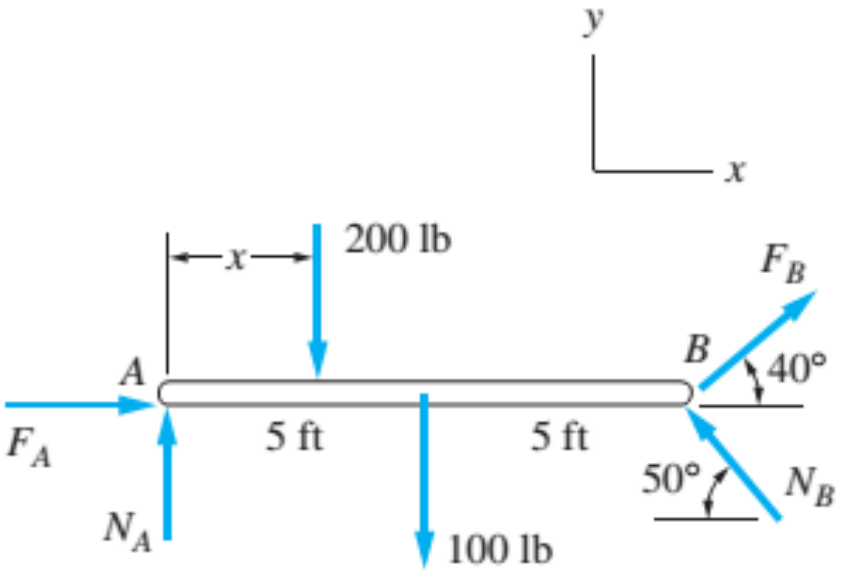
$\therefore$ Crate will not slide

FRICTION

- Example 7.4
- The uniform 100-lb plank is resting on friction surfaces at A and B. The coefficients of static friction are shown in the figure. If a 200-lb man starts walking from A toward B, determine the distance *x* when the plank will start to slide.



➢Example 7.4 – Solution



Assuming equilibriu m,

$$+\rightarrow \sum F_x = 0;\ F_A - (N_B \cos 50)\,\text{lb} + (F_B \cos 40)\,\text{lb} = 0$$

$$+\uparrow \sum F_y = 0;\ N_A - 200\ \text{lb} - 100\ \text{lb} + (N_B \sin 50)\,\text{lb} + (F_B \sin 40)\,\text{lb} = 0 F = -80\ \text{lb}$$

$$+\circlearrowleft \sum M_A = 0;\ (200\ \text{lb})x + (200\ \text{lb})5\ \text{ft} - (F_B \sin 40\ \text{lb})10\ \text{ft} - (N_B \sin 50\ \text{lb})10\ \text{ft} = 0$$

Subtitutin g $F_A = 0.2N_A$ and $F_B = 0.5N_B$ and solving,

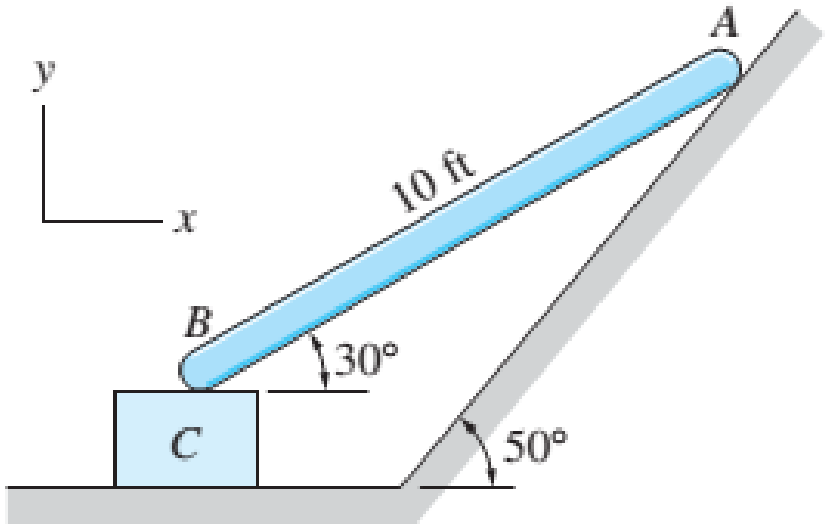
$N_A = 163.3\ \text{lb}$

$N_B = 125.7\ \text{lb}$

$x = 4.34\ \text{ft}$

FRICTION

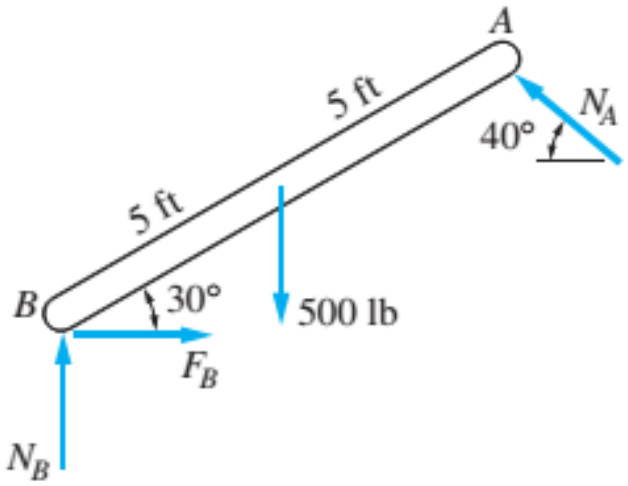
- Example 7.5
- Determine if the system below will be in static equilibrium. The uniform bar AB weighs 500 lb, and the weight of block C is 300 lb. Friction at A is negligible, and the coefficient of static friction is 0.4 at the other two contact surfaces.



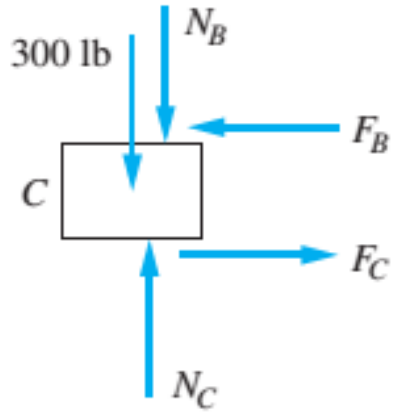
Ans:
The system will not be in equilibrium because F > F_{max} for the block, C.

FRICTION

➢Example 7.5-Solution



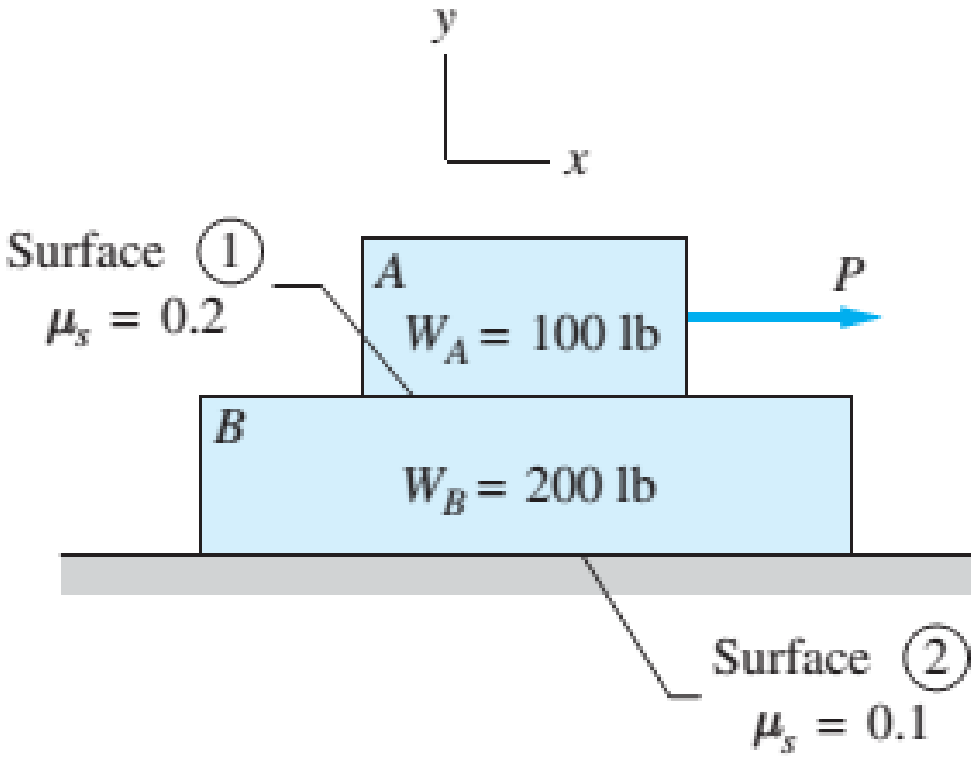
(b)



Ans:
The system will not be in equilibrium because F > F_{max} for the block, C.

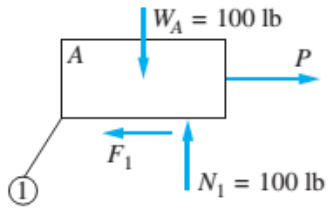
FRICTION

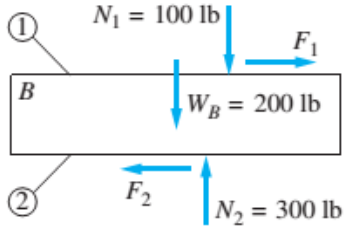
- Example 7.5
- Determine the maximum force P that can be applied to block A without causing either block to move.

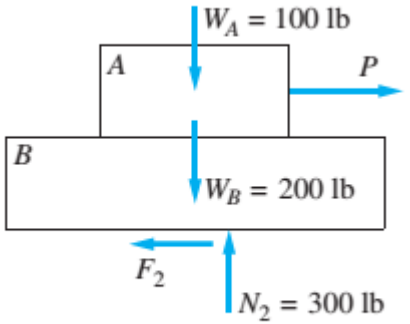


FRICTION

➢Example 7.5 - Solution







(b)

For surface 1,

$$F_1 = (F_1)_{\max} = (\mu_s)_1 N_1 = 0.2(100) = 20 \text{ lb}$$

The FBD of block A then gives

$$\Sigma F_x = 0 \quad \xrightarrow{+} \quad P - F_1 = 0$$
$$P = F_1 = 20 \text{ lb}$$

For surface 2,

$$F_2 = (F_2)_{\max} = (\mu_s)_2 N_2 = 0.1(300) = 30 \text{ lb}$$

From the FBD of the entire system, Fig. (b), we then obtain

$$\Sigma F_x = 0 \quad \xrightarrow{+} \quad P - F_2 = 0$$
$$P = F_2 = 30 \text{ lb}$$

Hence, the largest force that can be applied without causing motion is 20 lb

TAKE HOME ASSIGNMENT II